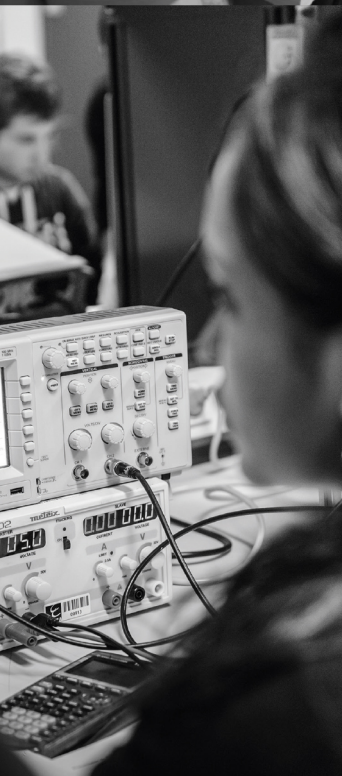




ENSEA

Beyond Engineering

POWER ELECTRONICS

DEE_2401

TD

Alexis MARTIN
2026

1 Powering a microprocessor

We want to power an STM32H4 microprocessor using a battery. The specifications are as follows:

- LiPo 2S battery $V_{in} = 7.4\text{ V}$
- Microprocessor supply voltage: $V_{out} = 3.3\text{ V}$
- Maximum voltage ripple at the microprocessor input: $\Delta V_S = 50\text{ mV}$
- Output nominal current: $I_S = 200\text{ mA}$
- Current ripple: 30%

To achieve this, we want to use the following component: **LT1959**, an excerpt of the documentation is attached. To size all the components, we decided to use IAs: Claude (Sonnet 4.6) and ChatGPT (GPT 5.5 Instant). The AI responses were generated in June 2026 in French and then translated in English (the original Prompt and responses are in the French version of this Tutorial). Here is the translation in English of the prompt used:

"I want to use the LT1959 component to power an STM32H4 using a battery. Here are the specifications:

- *LiPo 2S battery $V_{in} = 7.4\text{ V}$*
- *Microprocessor supply voltage: $V_{out} = 3.3\text{ V}$*
- *Maximum voltage ripple at the microprocessor input: $\Delta V_S = 50\text{ mV}$*
- *Output nominal current: $I_S = 200\text{ mA}$*
- *Current ripple: 30%*

In a Word file, detail the calculations for the sizing of passive components. Add an image of the electrical schematic. Add another image with the waveform of the current in the inductance, the voltage on the SW pin, and the output voltage".

With the AI responses and your knowledge:

1. Draw the electrical schematic of the Buck converter
2. Give the waveforms, in continuous conduction mode (CCM), of the voltage on the SW pin, the current in the inductor, and the output voltage (assuming that the voltage ripple comes mainly from the ESR of the capacitor)
3. Size the passive components of the Buck converter

Claude

Parameter	Value
Input voltage (LiPo 2S battery)	VIN = 7.4 V
Output voltage	VOUT = 3.3 V
Nominal output current	IS = 200 mA
Inductor current ripple	$\Delta IL = 30\%$ of IS
Max output voltage ripple	$\Delta VS = 50$ mV
Switching frequency (LT1959)	f = 500 kHz

1. Duty Cycle

For an ideal buck converter in continuous mode:

$$D = V_{OUT} / V_{IN}$$

$$D = 3,3 / 7,4$$

$$D = 0,446 \rightarrow t_{on} = 0,892 \mu s / t_{off} = 1,108 \mu s$$

Check : $D = 44,6 \% < 50 \% \rightarrow$ peak current $I_P = 4,5$ A guaranteed by LT1959.

2. Inductor L1

2.1 Target current ripple

The peak-to-peak current ripple in the inductor is set to 30% of the output current:

$$\Delta IL = 0,30 \times IS = 0,30 \times 200 \text{ mA}$$

$$\Delta IL = 60 \text{ mA}$$

2.2 Minimum value of L

The formula from the datasheet LT1959 for continuous conduction mode is:

$$L = (V_{OUT} \times (V_{IN} - V_{OUT})) / (V_{IN} \times \Delta IL \times f)$$

$$L = (3,3 \times (7,4 - 3,3)) / (7,4 \times 0,060 \times 500 \times 10^3)$$

$$L = (3,3 \times 4,1) / (7,4 \times 30\,000)$$

$$L = 13,53 / 222\,000$$

$$L_{min} = 60,9 \mu H$$

Standard value: $L1 = 10 \mu H$ (standard serie, available with $I_{sat} > 500$ mA).

Note: with $L = 10 \mu H$, the actual ripple will be lower than the 30% target:

$$\Delta IL_{réel} = (V_{OUT} \times (V_{IN} - V_{OUT})) / (V_{IN} \times L \times f)$$

$$\Delta IL_{réel} = (3,3 \times 4,1) / (7,4 \times 10 \times 10^{-6} \times 500 \times 10^3)$$

$$\Delta IL_{real} \approx 365 \text{ mA} \rightarrow 18 \% \text{ ok IS } \checkmark (< 30 \%)$$

3. peak and valley currents in the inductor

$$I_{L_peak} = IS + \Delta IL_{real}/2 = 200 + 183 = 383 \text{ mA}$$

$$I_{L_valley} = IS - \Delta IL_{real}/2 = 200 - 183 = 17 \text{ mA}$$

Continuous conduction mode check: $I_{L_valley} > 0 \checkmark$

Saturation current required: $I_{sat} > I_{L_peak} = 383 \text{ mA} \rightarrow$ inductor standard $10 \mu H$ ok.

4. Output capacitor C3

4.1 ESR constraint (voltage ripple)

The output voltage ripple is mainly determined by the capacitor's ESR:

$$\begin{aligned}\Delta V_S &\approx \Delta I_{L_real} \times ESR \\ ESR_{max} &= \Delta V_S / \Delta I_{L_real} = 50 \times 10^{-3} / 0,365 \\ ESR_{max} &= 136 \text{ m}\Omega\end{aligned}$$

Chosen capacitor: AVX TPS 100 μ F / 10 V (tantalum solide, ESR typ. 100 m Ω < 136 m Ω) ✓

4.2 Capacitance value

The value in μ F is not critical. With C3 = 100 μ F:

$$\begin{aligned}\text{Capacitive ripple} &= \Delta I_L / (8 \times f \times C3) \\ &= 0,365 / (8 \times 500 \times 10^3 \times 100 \times 10^{-6}) \\ &= 0,91 \text{ mV (negligible compared to the ESR contribution)}\end{aligned}$$

5. Input capacitor C1

The RMS current in the input capacitor is:

$$\begin{aligned}I_{C1_rms} &= I_S \times \sqrt{(V_{OUT} \times (V_{IN} - V_{OUT})) / V_{IN}} \\ I_{C1_rms} &= 0,200 \times \sqrt{(3,3 \times 4,1 / 7,4)} \\ I_{C1_rms} &= 0,200 \times \sqrt{0,183} \\ I_{C1_rms} &\approx 85,5 \text{ mA}\end{aligned}$$

Chosen capacitor: C1 = 100 μ F / 10 V, tantalum AVX TPS (Iripple $\geq$ 0,7 A) ✓

6. Freewheel diode D1

Average current in D1 :

$$\begin{aligned}ID1_{avg} &= I_S \times (V_{IN} - V_{OUT}) / V_{IN} \\ ID1_{avg} &= 0,200 \times (7,4 - 3,3) / 7,4 \\ ID1_{avg} &= 110 \text{ mA}\end{aligned}$$

Max reverse voltage = V_{IN} = 7,4 V

Selected Diode: MBRS330T3 (Schottky, 3 A / 30 V, V_F typ. 0,5 V @ 3 A) ✓

7. Capacitor BOOST C2

The capacitor BOOST supplies the switch NPN driver to saturate it. The minimum value is:

$$\begin{aligned}C2_{min} &= (I_S / 50) / (f \times 0,6) \\ C2_{min} &= (0,200 / 50) / (500 \times 10^3 \times 0,6) \\ C2_{min} &\approx 13 \text{ nF} \rightarrow C2 \text{ selected} = 0,68 \text{ }\mu\text{F (datasheet standard value)} \quad \checkmark\end{aligned}$$

Diode D2 : 1N914 or 1N4148, anode connected to VOUT.

8. Feedback network (R1, R2)

The internal voltage is V_{REF} = 1,21 V. The resistive divider R1/R2 fixes VOUT :

$$\begin{aligned}V_{OUT} &= V_{REF} \times (1 + R1/R2) \\ \rightarrow R1/R2 &= (V_{OUT}/V_{REF}) - 1 = (3,3/1,21) - 1 = 1,727\end{aligned}$$

Choice of R2 $\leq$ 2,5 k Ω (datasheet recommandation) : R2 = 2,49 k Ω

$$R1 = R2 \times (V_{OUT}/V_{REF} - 1) = 2490 \times 1,727$$

12. Simulated waveforms

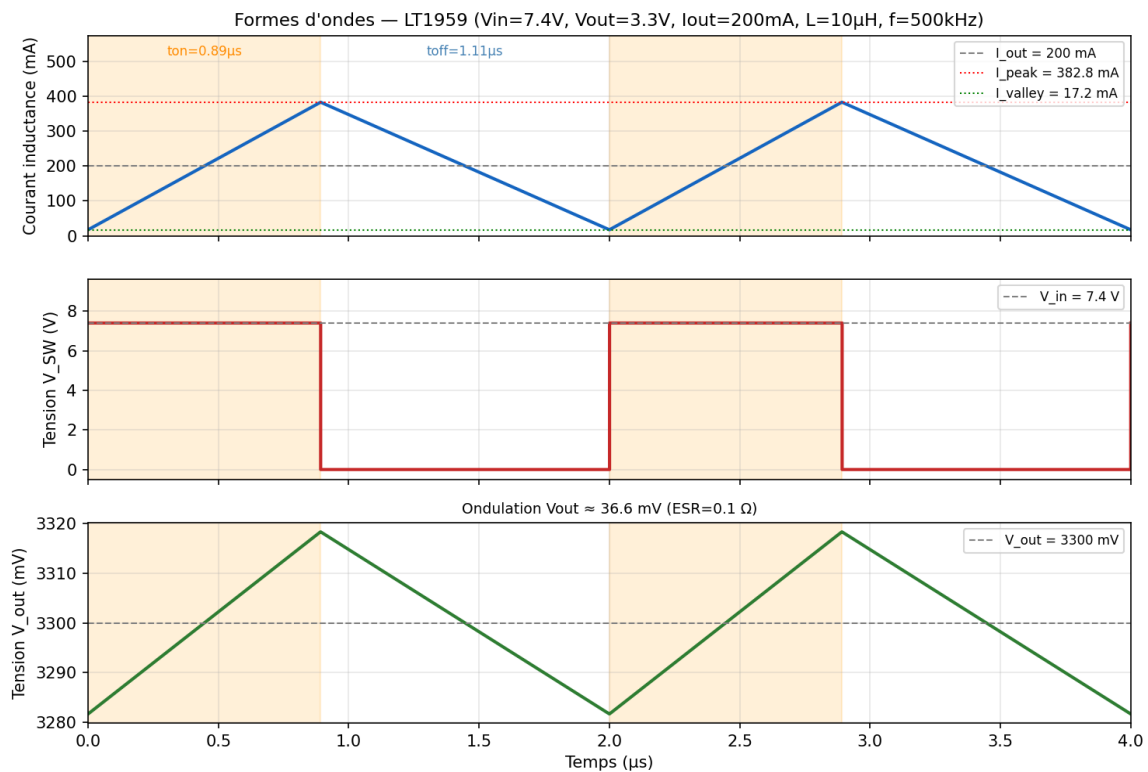


Figure 2 — Inductor current I_L , voltage V_{SW} and output voltage V_{OUT} over 2 periods ($f = 500 \text{ kHz}$)

The orange area represents the conduction time of the switch ($t_{on} \approx 0,892 \mu\text{s}$).

V_{OUT} ripple is mainly due to the ESR of the capacitor C3 ($\approx 36 \text{ mV} < 50 \text{ mV spec}$) ✓

Reference : Linear Technology — LT1959 Datasheet, Rev. A.

Calculus realized for $I_S = 200 \text{ mA}$ nominal. Check thermal parameters for higher currents.

ChatGPT

Purpose: To size the main passive components of an LT1959 buck converter to generate 3.3 V / 200 mA from a rated 7.4 V 2S LiPo battery, with a maximum requested output ripple of 50 mV and an inductor current ripple of 30%.

1. Data and assumptions

Size	Value	Comment
Vin	7.4 V	Rated Voltage 2S LiPo Battery
Vout	3.3 V	Microprocessor power supply
Is	200 mA	Rated Current
ΔVS max	50 mV peak-to-peak	Maximum Allowable Ripple at the STM32 Input
ΔIL	30% Is = 60 mA i.e.	Inductor Current Ripple
fs	500 kHz	LT1959 Fixed Frequency
VFB	1.21 V	FB Pin Reference

Important note: the calculations below are made at the ratted point Vin = 7.4 V. For a series design, it is also necessary to check the fully charged battery, typically 8.4 V for a 2S LiPo, and the dead battery. As the LT1959 is greatly oversized in current for 200 mA, the dominant constraints will be more ripple, stability, choice of output capacitor and routing.

2. Principle Electrical Diagram

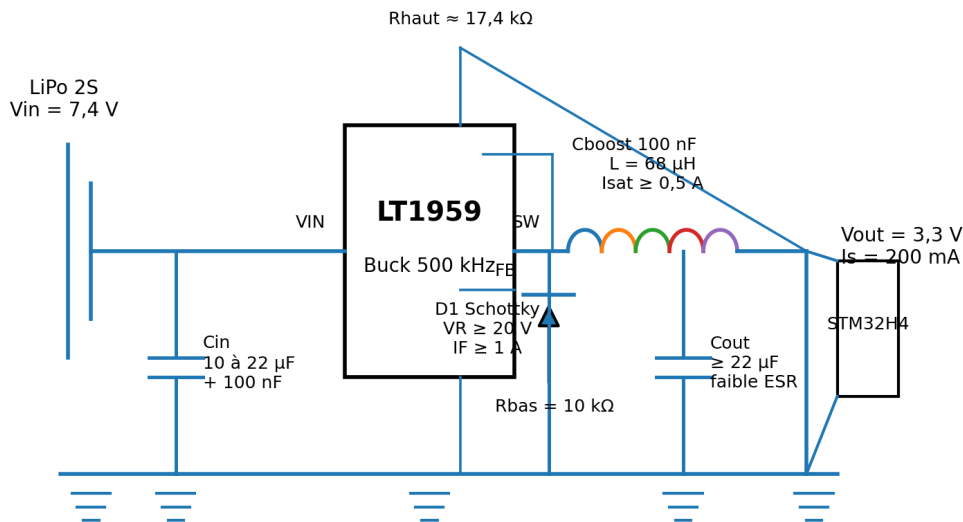


Schéma de principe pour le dimensionnement. Les valeurs finales doivent être validées avec la compensation, le routage et la fiche technique.

Figure 1 - Schematic diagram of the LT1959 to 3.3 V buck converter.

3. Duty cycle

As a first approximation, neglecting the drops in the internal switch and in the freewheel diode:

$D = V_{out} / V_{in} = 3.3 / 7.4 = 0.446$, i.e. 44.6%.

At 500 kHz, the switching period is $T = 1 / f_s = 2.00 \mu\text{s}$. We obtain $T_{on} = D.T = 0.89 \mu\text{s}$ and $T_{off} = (1-D). T = 1.11 \mu\text{s}$.

4. Inductor Sizing

For a buck in continuous conduction mode, the peak-to-peak ripple of the inductor current is:

$$\Delta I_L = (V_{in} - V_{out}). D / (L.f_s)$$

$$\text{So } L = (V_{in} - V_{out}). D / (f_s.\Delta I_L) = (7.4 - 3.3) \times 0.446 / (500 \text{ kHz} \times 60 \text{ mA}) = 60.9 \mu\text{H}.$$

Recommended normalized value: 68 μH . With 68 μH , the ripple is about 53.8 mA p-to-p, or about 26.9% of the rated current.

Peak current in the inductor: $I_{pk} = I_s + \Delta I_L/2 = 200 \text{ mA} + 30 \text{ mA} = 230 \text{ mA}$. Minimum current: $I_{min} = 170 \text{ mA}$. Choose an inductor with a much higher saturation current, e.g. $I_{sat} \geq 0.5 \text{ A}$, and a low series resistor to limit losses at low voltages.

5. Output capacitor

The output ripple has two main contributions: capacitive load/discharge and ESR drop of the capacitor. The ideal capacitive component is:

$$\Delta V_{cap} = \Delta I_L / (8.f_s. C_{out})$$

$$\text{Imposing } \Delta V_{cap} \leq 50 \text{ mV: } C_{out} \geq \Delta I_L / (8.f_s.\Delta V) = 60 \text{ mA} / (8 \times 500 \text{ kHz} \times 50 \text{ mV}) = 0.30 \mu\text{F}.$$

The purely capacitive stress is therefore very low. The ESR constraint gives: $ESR \leq \Delta V / \Delta I_L = 50 \text{ mV} / 60 \text{ mA} = 0.83 \Omega$.

Recommended practical choice: at least 22 μF ceramic X7R in the output, supplemented if necessary by 100 nF close to the load. The actual value of a ceramic capacitor decreases with DC polarization, so the effective capacitance at 3.3 V must be checked. For a microprocessor, the goal is not only ripple at 500 kHz, but also to withstand fast current demands.

6. Input capacitor

The input capacitor must provide the pulsed current from the converter and remain close to the VIN/GND pins of the LT1959. Its approximate RMS current is:

$$I_{cin,rms} \approx I_s.\sqrt{D.(1-D)} = 0.2 \times \sqrt{0.446 \times 0.554} = 99 \text{ mA RMS}.$$

To aim, for example, for 100 mV of low-frequency switching input ripple, an estimate gives $C_{in} \geq I_s.D. (1-D)/(f_s.\Delta V_{in}) = 0.99 \mu\text{F}$. In practice, retain 10 to 22 μF low ESR, with a ceramic 100 nF directly as close as possible to VIN and GND.

7. FB Return Bridge

The output voltage is set by: $V_{out} = V_{FB} \times (1 + R_{haut}/R_{bas})$.

$$\text{By choosing } R_{bas} = 10 \text{ k}\Omega: R_{haut} = 10 \text{ k}\Omega \times (3.3/1.21 - 1) = 17.27 \text{ k}\Omega.$$

Normalized value: $R_{haut} = 17.4 \text{ k}\Omega$ and $R_{bas} = 10 \text{ k}\Omega$, which gives $V_{out} \approx 3.32 \text{ V}$. An alternative is $R_{haut} = 17.2 \text{ k}\Omega$ to obtain about 3.29 V.

8. Freewheel Diode and Boost Capacitor

The freewheel LED drives during Toff. Its average current is $I_{d,avg} \approx I_s \cdot (1-D) = 0.2 \times 0.554 = 111 \text{ mA}$. Its peak current is close to the inductance peak current, i.e. about 230 mA at the rated point.

Choose a fast Schottky diode with 20 V VRRM and $I_F \geq 1$ for a comfortable headroom. The reverse voltage must at least support V_{in} , and the routing must minimize the LT1959 loop - diode - input capacitor.

The boost capacitor used by the internal control is typically 100 nF, placed as close as possible to the BOOST and SW pins, according to the application diagram of the data sheet.

9. Expected Waveforms

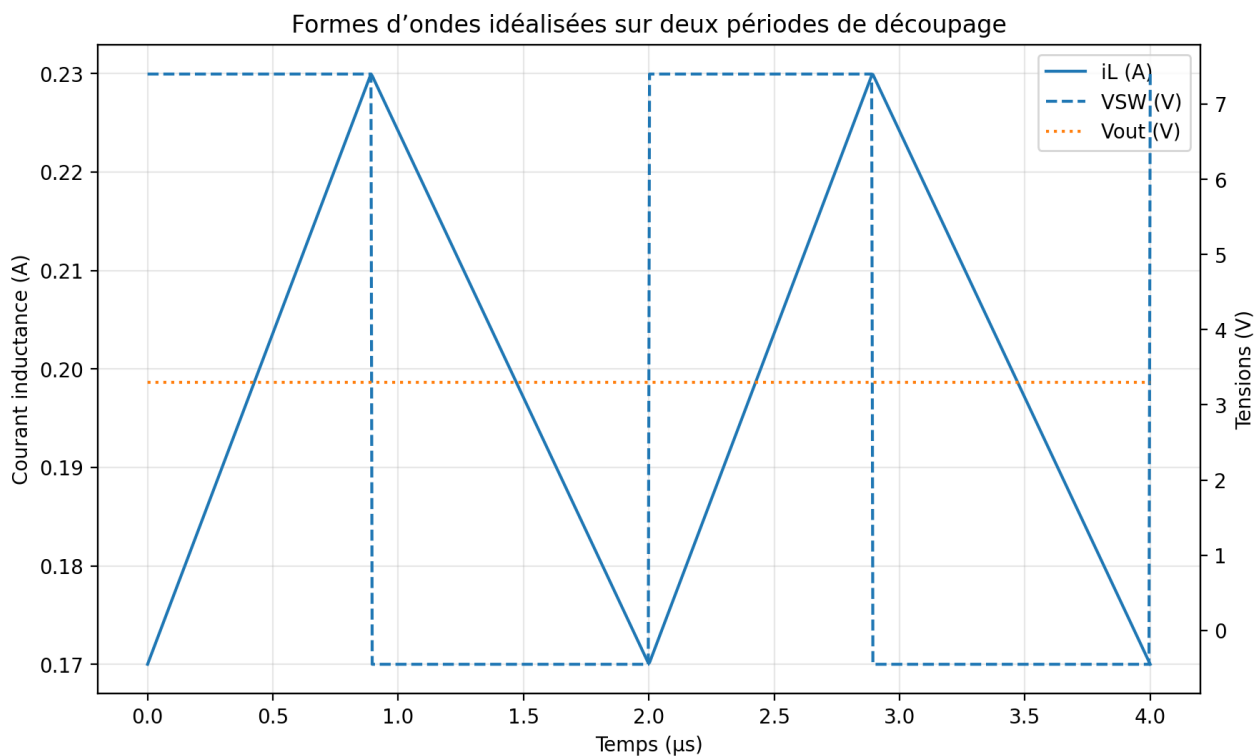


Figure 2 - Current in the inductor, SW voltage and output ripple idealized.

The inductance current is triangular in continuous conduction. The SW pin switches between a value close to V_{in} during Ton and a slightly negative voltage during Toff, limited by the freewheel diode. The output voltage is almost continuous, with residual ripple dependent on ΔI_L , Cout, ESR and routing.

10. Summary of the proposed values

Component	Calculated Value / Constraint	Proposed choice
L	60.9 μ H for $\Delta I_L = 60$ mA	68 μ H, $I_{sat} \geq 0.5$ A
Cout	≥ 0.30 μ F and $ESR \leq 0.83$ Ω	22 μ F X7R + 100 nF, low ESR
Cin	$I_{rms} \approx 99$ mA; Low minimum capacity	10 to 22 μ F low ESR + 100 nF near LT1959
Rhaut/Rbas	$R_{haut} = 17.27$ k Ω as $R_{bas} = 10$ k Ω	17.4 k Ω / 10 k Ω
D1	$I_{d,avg} \approx 111$ mA; Peak ≈ 230 mA	Schottky ≥ 20 V, ≥ 1 A
Cboost	According to LT1959 application	100 nF near BOOST-SW

Conclusion: At the rated point 7.4 V to 3.3 V / 200 mA, a 68 μ H inductance and a 22 μ F ceramic output capacitor give a comfortable margin with respect to the 50 mV ripple. However, loop stability must be validated with VC pin compensation, actual capacitor characteristics, and PCB routing.

References

Analog Devices, LT1959 product page and data sheet: 500 kHz buck regulator, 1.21 V FB adjustable output, 4.5 A switch, 4 V minimum input, VIN/GND/SW/diode routing recommendations.

4.5A, 500kHz Step-Down Switching Regulator

FEATURES

- Operates with Input as Low as 4V
- Output Range Down to 1.21V
- Constant 500kHz Switching Frequency
- Uses All Surface Mount Components
- Inductor Size Reduced to 1.8μH
- Saturating Switch Design: 0.07Ω
- Shutdown Current: 15μA
- Easily Synchronizable
- Cycle-by-Cycle Current Limiting
- 4.5A Switch
- Current Mode Control

APPLICATIONS

- Portable Computers
- Battery-Powered Systems
- Battery Chargers
- Distributed Power
- 5V to 3.3V Conversion
- 5V to 2.5V Conversion
- 5V to 1.8V Conversion

DESCRIPTION

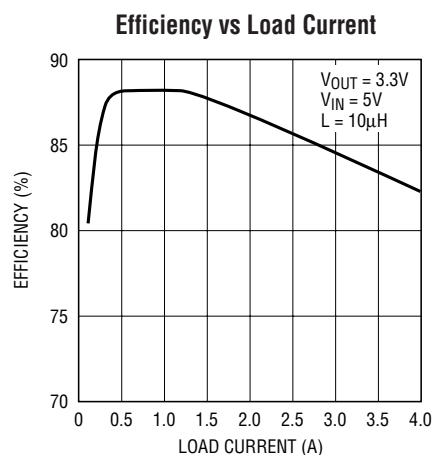
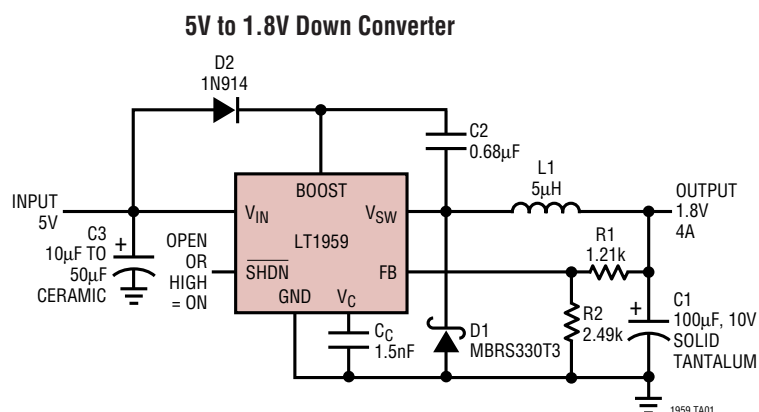
The LT[®]1959 is a 500kHz monolithic buck mode switching regulator functionally identical to the LT1506 but optimized for lower output voltage applications. It will operate down to 1.21V output compared to 2.42V for the LT1506. A 4.5A switch is included on the die along with all the necessary oscillator, control and logic circuitry. High switching frequency allows a considerable reduction in the size of external components. The topology is current mode for fast transient response and good loop stability.

A special high speed bipolar process and new design techniques achieve high efficiency at high switching frequency. Efficiency is maintained over a wide output current range by keeping quiescent supply current to 3.8mA and by utilizing a supply boost capacitor to saturate the power switch.

The LT1959 fits into standard 7-pin DD and fused lead SO-8 packages. Full cycle-by-cycle short-circuit protection and thermal shutdown are provided. Standard surface mount external parts are used, including the inductor and capacitors. There is the optional function of shutdown or synchronization. A shutdown signal reduces supply current to 15μA. Synchronization allows an external logic level signal to increase the internal oscillator from 580kHz to 1MHz.

LT, LTC and LT are registered trademarks of Linear Technology Corporation.

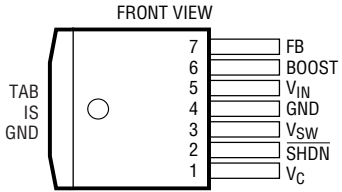
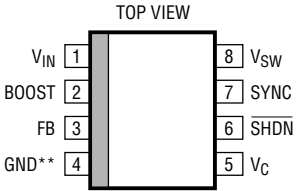
TYPICAL APPLICATION



ABSOLUTE MAXIMUM RATINGS (Note 1)

Input Voltage	16V	SYNC Pin Voltage	7V
BOOST Voltage	30V	Operating Junction Temperature Range	
BOOST Pin Above Input Voltage	15V	LT1959C	0°C to 125°C
SHDN Pin Voltage	7V	LT1959I	–40°C to 125°C
FB Pin Voltage	3.5V	Storage Temperature Range	–65°C to 150°C
FB Pin Current	1mA	Lead Temperature (Soldering, 10 sec)	300°C

PACKAGE/ORDER INFORMATION

 FRONT VIEW R PACKAGE 7-LEAD PLASTIC DD $T_{JMAX} = 150^{\circ}\text{C}$, $\theta_{JA} = 30^{\circ}\text{C/W}$ WITH PACKAGE SOLDERED TO 0.5 SQUARE INCH COPPER AREA OVER BACKSIDE GROUND PLANE OR INTERNAL POWER PLANE. θ_{JA} CAN VARY FROM 20°C/W TO >40°C/W DEPENDING ON MOUNTING TECHNIQUES	ORDER PART NUMBER	 TOP VIEW S8 PACKAGE 8-LEAD PLASTIC SO $T_{JMAX} = 150^{\circ}\text{C}$, $\theta_{JA} = 80^{\circ}\text{C/W}$ ** WITH (FUSED GND) GROUND PIN CONNECTED TO GROUND PLANE OR LARGE LANDS	ORDER PART NUMBER
	LT1959CR LT1959IR		LT1959CS8 LT1959IS8
			S8 PART MARKING
			1959 1959I

ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_A = 25^{\circ}\text{C}$, $T_J = 25^{\circ}\text{C}$, $V_{IN} = 5\text{V}$, $V_C = 1.5\text{V}$, $\text{Boost} = V_{IN} + 5\text{V}$, switch open, unless otherwise noted.

PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
Feedback Voltage (Adjustable)	All Conditions	●	1.19	1.21	1.23	V
Reference Voltage Line Regulation	$4.3\text{V} \leq V_{IN} \leq 15\text{V}$			0.01	0.03	%/V
Feedback Input Bias Current		●	–0.5	0	0.5	μA
Error Amplifier Voltage Gain	(Note 2)		200	400		
Error Amplifier Transconductance	$\Delta I(V_C) = \pm 10\mu\text{A}$	●	1500 1000	2000	2700 3100	μMho μMho
V_C Pin to Switch Current Transconductance				5.3		A/V
Error Amplifier Source Current	$V_{FB} = 1.05\text{V}$	●	140	225	320	μA
Error Amplifier Sink Current	$V_{FB} = 1.35\text{V}$	●	140	225	320	μA
V_C Pin Switching Threshold	Duty Cycle = 0			0.9		V
V_C Pin High Clamp				2.1		V
Switch Current Limit	V_C Open, $V_{FB} = 1.05\text{V}$, DC $\leq 50\%$	●	4.5	6	8.5	A
Slope Compensation	DC = 80%			0.8		A

ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_A = 25^\circ\text{C}$, $T_J = 25^\circ\text{C}$, $V_{IN} = 5\text{V}$, $V_C = 1.5\text{V}$, $\text{Boost} = V_{IN} + 5\text{V}$, switch open, unless otherwise noted.

PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
Switch On Resistance (Note 7)	$I_{SW} = 4.5\text{A}$	●		0.07	0.1 0.13	Ω Ω
Maximum Switch Duty Cycle	$V_{FB} = 1.05\text{V}$	●	90 86	93 93		% %
Switch Frequency	V_C Set to Give 50% Duty Cycle	●	460 440	500	540 560	kHz kHz
Switch Frequency Line Regulation	$4.3\text{V} \leq V_{IN} \leq 15\text{V}$	●		0	0.15	%/V
Frequency Shifting Threshold on FB Pin	$\Delta f = 10\text{kHz}$	●	0.5	0.7	1.0	V
Minimum Input Voltage (Note 3)		●		4.0	4.3	V
Minimum Boost Voltage (Note 4)	$I_{SW} \leq 4.5\text{A}$	●		2.3	3.0	V
Boost Current (Note 5)	$I_{SW} = 1\text{A}$	●		20	35	mA
	$I_{SW} = 4.5\text{A}$	●		90	140	mA
Input Supply Current (Note 6)		●		3.8	5.4	mA
Shutdown Supply Current	$V_{SHDN} = 0\text{V}$, $V_{SW} = 0\text{V}$, V_C Open	●		15	50 75	μA μA
Lockout Threshold	V_C Open	●	2.3	2.38	2.46	V
Shutdown Thresholds	V_C Open Device Shutting Down	●	0.13	0.37	0.60	V
	Device Starting Up	●	0.25	0.45	0.7	V
Synchronization Threshold		●		1.5	2.2	V
Synchronizing Range			580		1000	kHz
SYNC Pin Input Resistance				40		$\text{k}\Omega$

Note 1: Absolute Maximum Ratings are those values beyond which the life of a device may be impaired.

Note 2: Gain is measured with a V_C swing equal to 200mV above the switching threshold level to 200mV below the upper clamp level.

Note 3: Minimum input voltage is not measured directly, but is guaranteed by other tests. It is defined as the voltage where internal bias lines are still regulated so that the reference voltage and oscillator frequency remain constant. Actual minimum input voltage to maintain a regulated output will depend on output voltage and load current. See Applications Information.

Note 4: This is the minimum voltage across the boost capacitor needed to guarantee full saturation of the internal power switch.

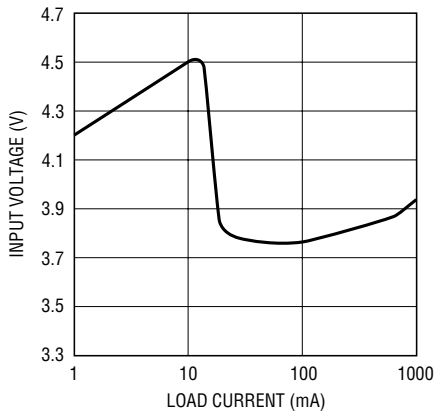
Note 5: Boost current is the current flowing into the boost pin with the pin held 5V above input voltage. It flows only during switch on time.

Note 6: Input supply current is the bias current drawn by the input pin with switching disabled.

Note 7: Switch on resistance is calculated by dividing V_{IN} to V_{SW} voltage by the forced current (4.5A). See Typical Performance Characteristics for the graph of switch voltage at other currents.

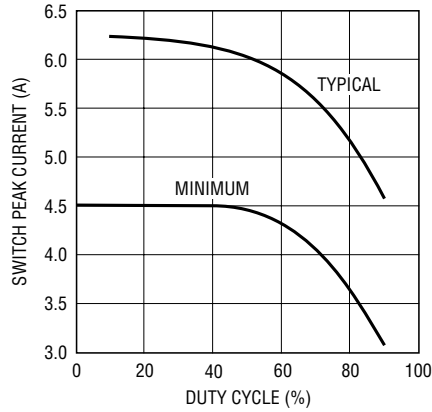
TYPICAL PERFORMANCE CHARACTERISTICS

**Minimum Input Voltage
with 3.3V Output**



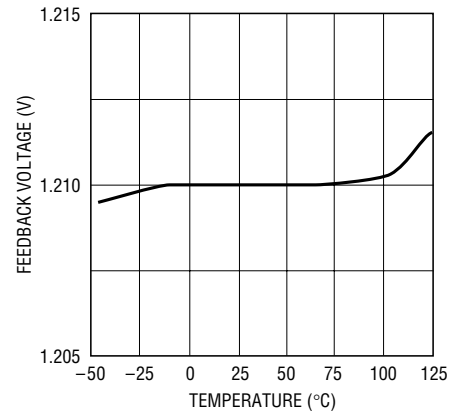
1959 G01

Switch Peak Current Limit



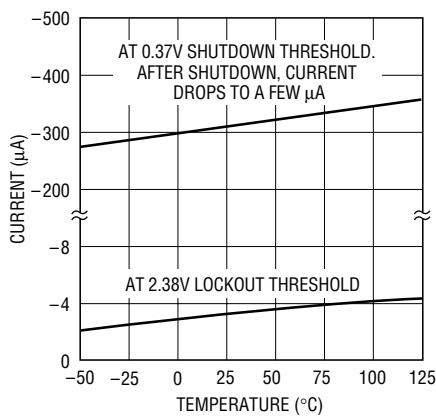
1959 G02

Feedback Pin Voltage



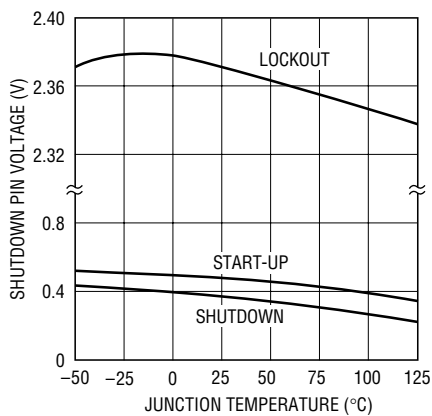
1959 G03

Shutdown Pin Bias Current



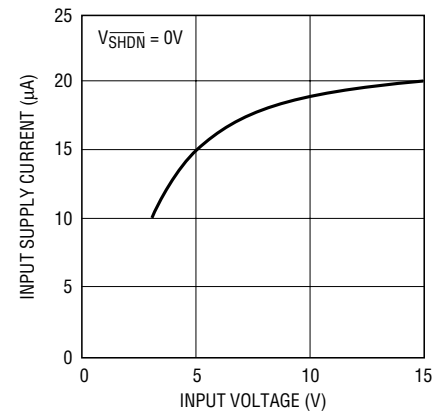
1959 G04

**Lockout and Shutdown
Thresholds**



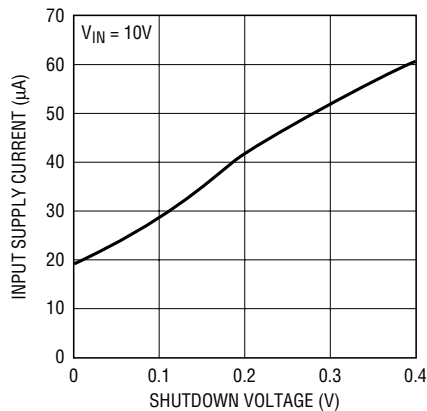
1959 G05

Shutdown Supply Current



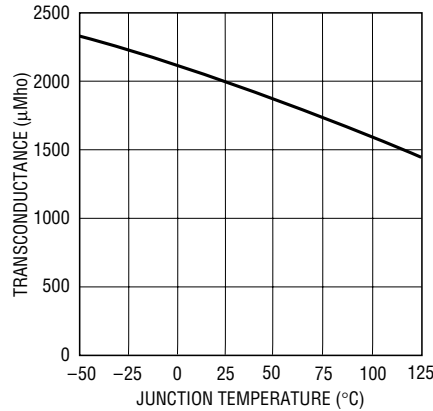
1959 G06

Shutdown Supply Current



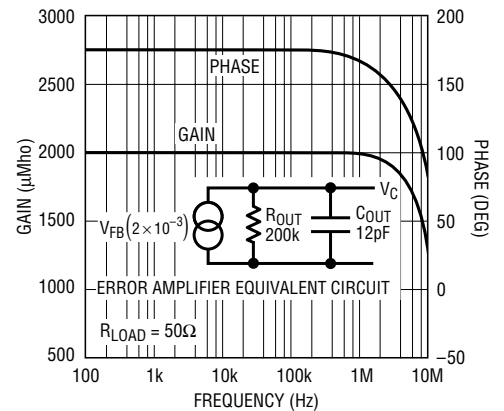
1959 G07

Error Amplifier Transconductance



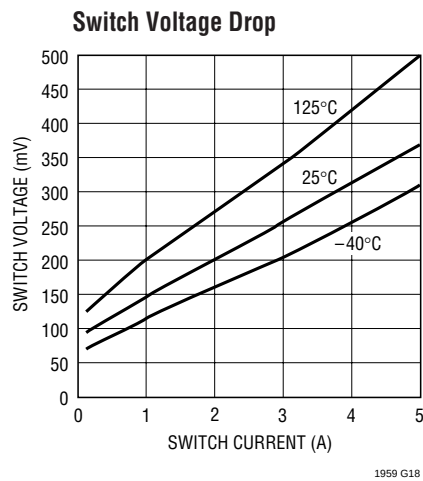
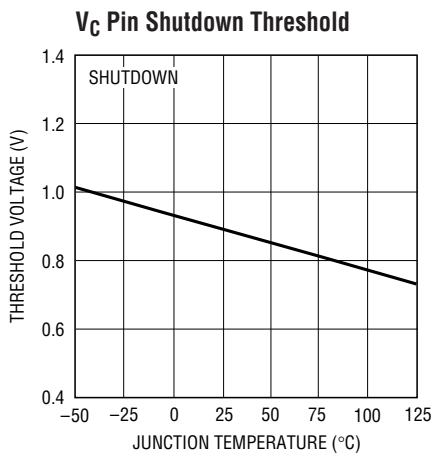
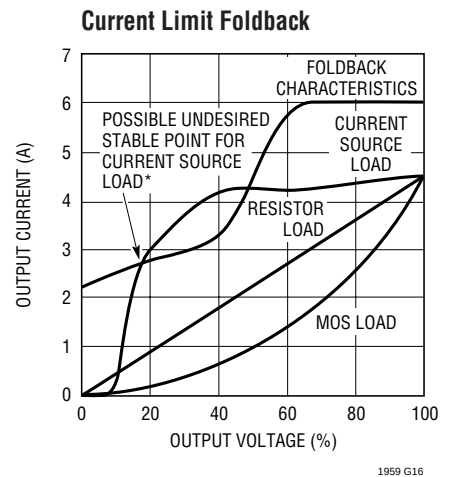
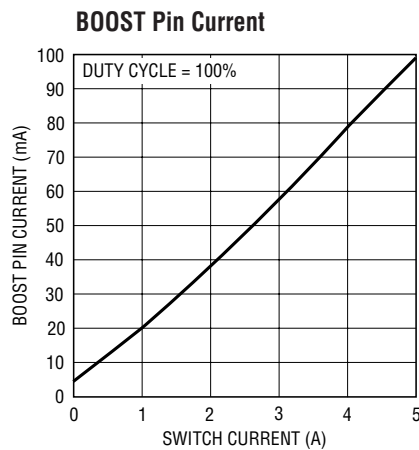
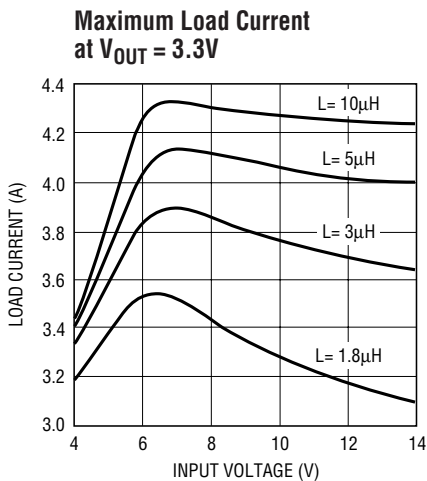
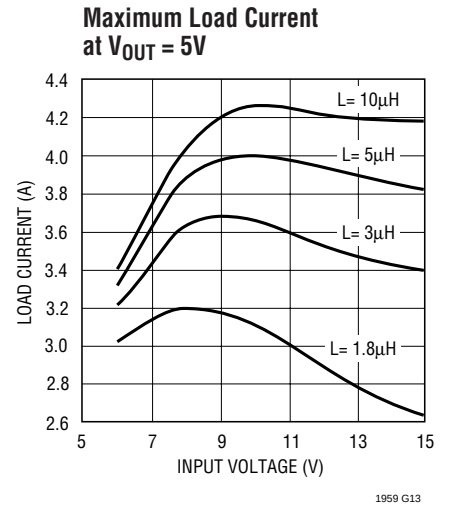
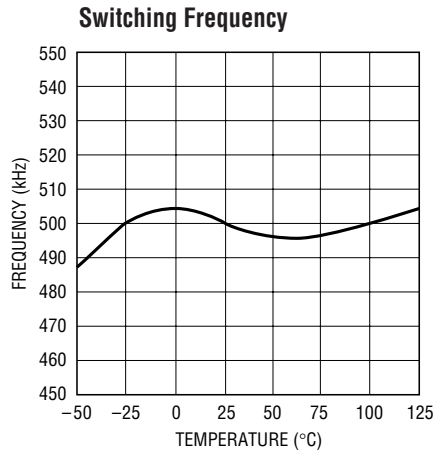
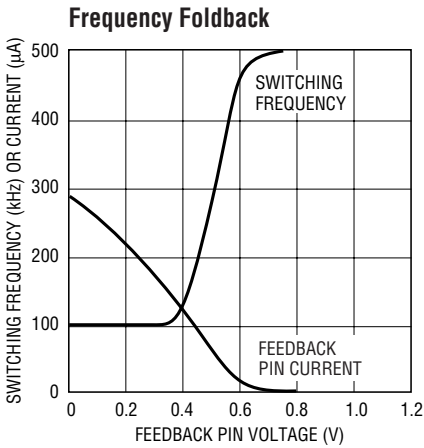
1959 G08

Error Amplifier Transconductance



1959 G09

TYPICAL PERFORMANCE CHARACTERISTICS



*See "More Than Just Voltage Feedback" in the Applications Information section.

PIN FUNCTIONS

FB: The feedback pin is used to set output voltage using an external voltage divider that generates 1.21V at the pin with the desired output voltage. Three additional functions are performed by the FB pin. When the pin voltage drops below 0.8V, switch current limit is reduced. Below 0.7V the external sync function is disabled and switching frequency is reduced. See Feedback Pin Function section in Applications Information for details.

BOOST: The BOOST pin is used to provide a drive voltage, higher than the input voltage, to the internal bipolar NPN power switch. Without this added voltage, the typical switch voltage loss would be about 1.5V. The additional boost voltage allows the switch to saturate and voltage loss approximates that of a 0.07Ω FET structure, but with much smaller die area. Efficiency improves from 75% for conventional bipolar designs to > 89% for these new parts.

V_{IN}: This is the collector of the on-chip power NPN switch. This pin powers the internal circuitry and internal regulator. At NPN switch on and off, high dI/dt edges occur on this pin. Keep the external bypass and catch diode close to this pin. All trace inductance on this path will create a voltage spike at switch off, adding to the V_{CE} voltage across the internal NPN.

GND: The GND pin connection needs consideration for two reasons. First, it acts as the reference for the regulated output, so load regulation will suffer if the “ground” end of the load is not at the same voltage as the GND pin of the IC. This condition will occur when load current or other currents flow through metal paths between the GND pin and the load ground point. Keep the ground path short between the GND pin and the load and use a ground plane when possible. The second consideration is EMI caused by GND pin current spikes. Internal capacitance between the V_{SW} pin and the GND pin creates very narrow (<10ns) current spikes in the GND pin. If the GND pin is connected to system ground with a long metal trace, this trace may

radiate excess EMI. Keep the path between the input bypass and the GND pin short. The GND pin of the SO-8 package is directly attached to the internal tab. This pin should be attached to a large copper area to improve thermal resistance.

V_{SW}: The switch pin is the emitter of the on-chip power NPN switch. This pin is driven up to the input pin voltage during switch on time. Inductor current drives the switch pin negative during switch off time. Negative voltage is clamped with the external catch diode. Maximum negative switch voltage allowed is -0.8V.

SYNC: (SO8 Package Only) The sync pin is used to synchronize the internal oscillator to an external signal. It is directly logic compatible and can be driven with any signal between 10% and 90% duty cycle. The synchronizing range is equal to *initial* operating frequency, up to 1MHz. See Synchronizing section in Applications Information for details. When not in use, this pin should be grounded.

SHDN: The shutdown pin is used to turn off the regulator and to reduce input drain current to a few microamperes. Actually, this pin has two separate thresholds, one at 2.38V to disable switching, and a second at 0.4V to force complete micropower shutdown. The 2.38V threshold functions as an accurate undervoltage lockout (UVLO). This is sometimes used to prevent the regulator from operating until the input voltage has reached a predetermined level.

V_C: The V_C pin is the output of the error amplifier and the input of the peak switch current comparator. It is normally used for frequency compensation, but can do double duty as a current clamp or control loop override. This pin sits at about 1V for very light loads and 2V at maximum load. It can be driven to ground to shut off the regulator, but if driven high, current must be limited to 4mA.

BLOCK DIAGRAM

The LT1959 is a constant frequency, current mode buck converter. This means that there is an internal clock and two feedback loops that control the duty cycle of the power switch. In addition to the normal error amplifier, there is a current sense amplifier that monitors switch current on a cycle-by-cycle basis. A switch cycle starts with an oscillator pulse which sets the R_S flip-flop to turn the switch on. When switch current reaches a level set by the inverting input of the comparator, the flip-flop is reset and the switch turns off. Output voltage control is obtained by using the output of the error amplifier to set the switch current trip point. This technique means that the error amplifier commands current to be delivered to the output rather than voltage. A voltage fed system will have low phase shift up to the resonant frequency of the inductor

and output capacitor, then an abrupt 180° shift will occur. The current fed system will have 90° phase shift at a much lower frequency, but will not have the additional 90° shift until well beyond the LC resonant frequency. This makes it much easier to frequency compensate the feedback loop and also gives much quicker transient response.

High switch efficiency is attained by using the BOOST pin to provide a voltage to the switch driver which is higher than the input voltage, allowing switch to be saturated. This boosted voltage is generated with an external capacitor and diode. Two comparators are connected to the shutdown pin. One has a 2.38V threshold for undervoltage lockout and the second has a 0.4V threshold for complete shutdown.

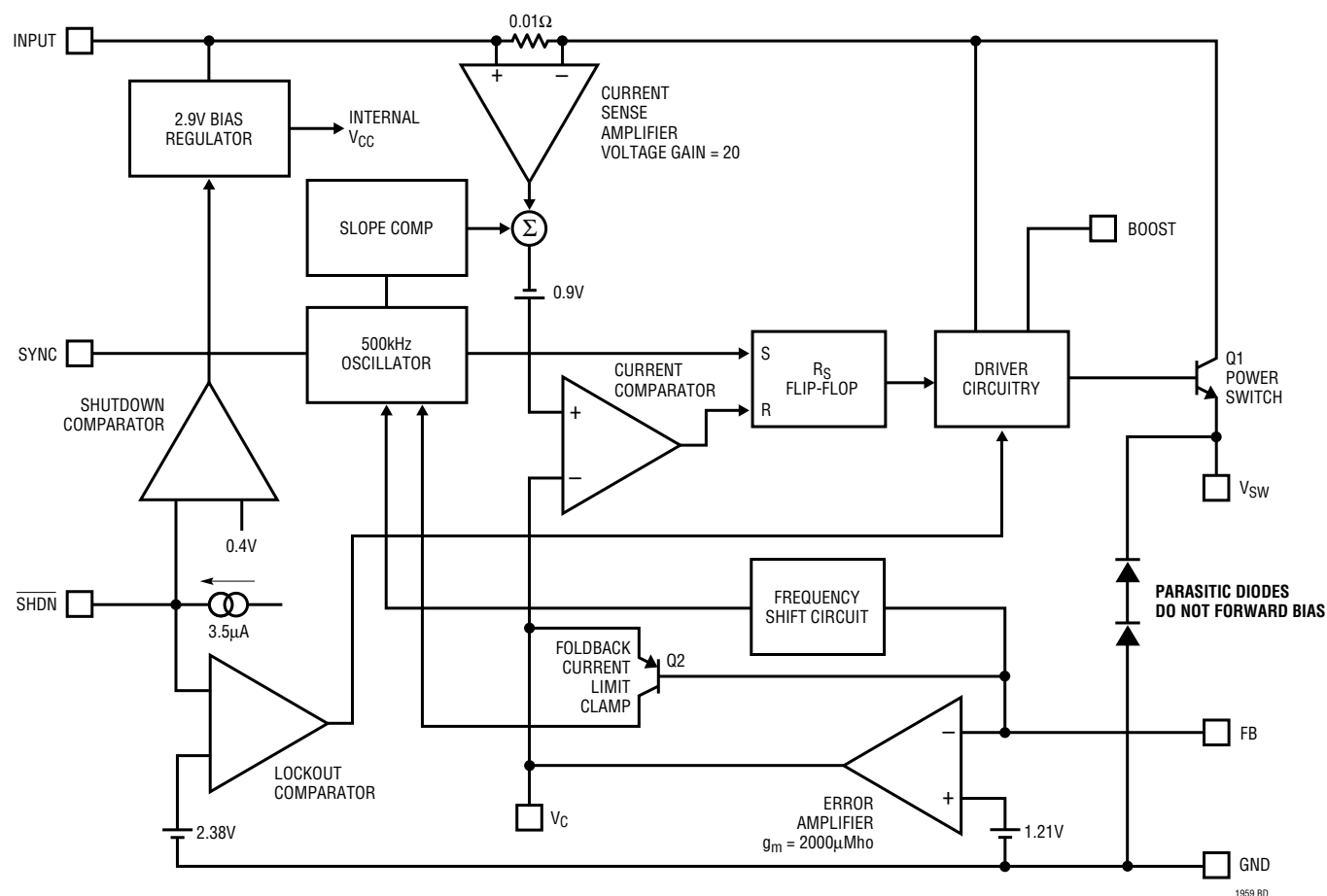


Figure 1. Block Diagram

APPLICATIONS INFORMATION

FEEDBACK PIN FUNCTIONS

The feedback (FB) pin on the LT1959 is used to set output voltage and provide several overload protection features. The first part of this section deals with selecting resistors to set output voltage and the remaining part talks about foldback frequency and current limiting created by the FB pin. Please read both parts before committing to a final design.

The suggested value for the output divider resistor (see Figure 2) from FB to ground (R2) is 2.5k or less, and a formula for R1 is shown below. The output voltage error caused by ignoring the input bias current on the FB pin is less than 0.1% with R2 = 2.5k. Please read the following if divider resistors are increased above the suggested values.

$$R1 = \frac{R2(V_{OUT} - 1.21)}{1.21}$$

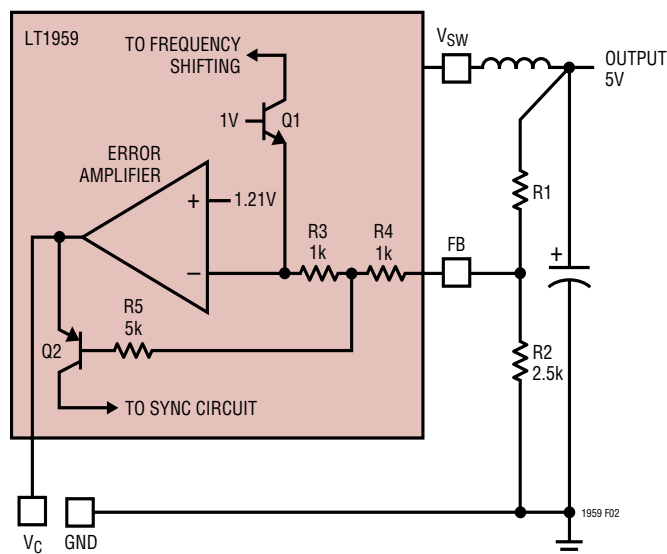


Figure 2. Frequency and Current Limit Foldback

More Than Just Voltage Feedback

The feedback pin is used for more than just output voltage sensing. It also reduces switching frequency and current limit when output voltage is very low (see the Frequency Foldback graph in Typical Performance Characteristics). This is done to control power dissipation in both the IC and in the external diode and inductor during short-circuit conditions. A shorted output requires the switching regulator to operate at very low duty cycles, and the average current through the diode and inductor is equal to the short-circuit current limit of the switch (typically 6A for the LT1959, folding back to less than 3A). Minimum switch on time limitations would prevent the switcher from attaining a sufficiently low duty cycle if switching frequency were maintained at 500kHz, so frequency is reduced by about 5:1 when the feedback pin voltage drops below 0.5V (see Frequency Foldback graph). This does not affect operation with normal load conditions; one simply sees a gear shift in switching frequency during start-up as the output voltage rises.

In addition to lower switching frequency, the LT1959 also operates at lower switch current limit when the feedback pin voltage drops below 0.8V. Q2 in Figure 2 performs this function by clamping the V_C pin to a voltage less than its normal 2.1V upper clamp level. This *foldback current limit* greatly reduces power dissipation in the IC, diode and inductor during short-circuit conditions. External synchronization is also disabled to prevent interference with foldback operation. Again, it is nearly transparent to the user under normal load conditions. The only loads that may be affected are current source loads which maintain full load current with output voltage less than 50% of final value. In these rare situations the feedback pin can be clamped above 0.75V with an external diode to defeat foldback current limit. **Caution:** clamping the feedback pin means that frequency shifting will also be defeated, so a combination of high input voltage and dead shorted output may cause the LT1959 to lose control of current limit.

APPLICATIONS INFORMATION

The internal circuitry which forces reduced switching frequency also causes current to flow out of the feedback pin when output voltage is low. The equivalent circuitry is shown in Figure 2. Q1 is completely off during normal operation. If the FB pin falls below 0.7V, Q1 begins to conduct current and reduces frequency at the rate of approximately 2kHz/μA. To ensure adequate frequency foldback (under worst-case short-circuit conditions), the external divider Thevinin resistance must be low enough to pull 150μA out of the FB pin with 0.3V on the pin ($R_{DIV} \leq 2k$). *The net result is that reductions in frequency and current limit are affected by output voltage divider impedance. Although divider impedance is not critical, caution should be used if resistors are increased beyond the suggested values and short-circuit conditions will occur with high input voltage.* High frequency pickup will increase and the protection accorded by frequency and current foldback will decrease.

MAXIMUM OUTPUT LOAD CURRENT

Maximum load current for a buck converter is limited by the maximum switch current rating (I_P) of the LT1959. This current rating is 4.5A up to 50% duty cycle (DC), decreasing to 3.7A at 80% duty cycle. This is shown graphically in Typical Performance Characteristics and as shown in the formula below:

$$I_P = 4.5A \text{ for } DC \leq 50\%$$

$$I_P = 3.21 + 5.95(DC) - 6.75(DC)^2 \text{ for } 50\% < DC < 90\%$$

$$DC = \text{Duty cycle} = V_{OUT}/V_{IN}$$

Example: with $V_{OUT} = 5V$, $V_{IN} = 8V$; $DC = 5/8 = 0.625$, and;

$$I_{SW(MAX)} = 3.21 + 5.95(0.625) - 6.75(0.625)^2 = 4.3A$$

Current rating decreases with duty cycle because the LT1959 has internal slope compensation to prevent current mode subharmonic switching. For more details, read Application Note 19. The LT1959 is a little unusual in this regard because it has nonlinear slope compensation which gives better compensation with less reduction in current limit.

Maximum load current would be equal to maximum switch current *for an infinitely large inductor*, but with

finite inductor size, maximum load current is reduced by one-half peak-to-peak inductor current. The following formula assumes continuous mode operation, implying that the term on the right is less than one-half of I_P .

$$I_{OUT(MAX)} = I_P - \frac{(V_{OUT})(V_{IN} - V_{OUT})}{2(L)(f)(V_{IN})}$$

For the conditions above and $L = 3.3\mu H$,

$$\begin{aligned} I_{OUT(MAX)} &= 4.3 - \frac{(5)(8-5)}{2(3.3 \cdot 10^{-6})(500 \cdot 10^3)(8)} \\ &= 4.3 - 0.57 = 3.73A \end{aligned}$$

At $V_{IN} = 15V$, duty cycle is 33%, so I_P is just equal to a fixed 4.5A, and $I_{OUT(MAX)}$ is equal to:

$$\begin{aligned} &4.5 - \frac{(5)(15-5)}{2(3.3 \cdot 10^{-6})(500 \cdot 10^3)(15)} \\ &= 4.5 - 1.01 = 3.49A \end{aligned}$$

Note that there is less load current available at the higher input voltage because inductor ripple current increases. This is not always the case. Certain combinations of inductor value and input voltage range may yield lower available load current at the lowest input voltage due to reduced peak switch current at high duty cycles. If load current is close to the maximum available, please check maximum available current at both input voltage extremes. To calculate actual peak switch current with a given set of conditions, use:

$$I_{SW(PEAK)} = I_{OUT} + \frac{V_{OUT}(V_{IN} - V_{OUT})}{2(L)(f)(V_{IN})}$$

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CHOOSING THE INDUCTOR AND OUTPUT CAPACITOR

For most applications the output inductor will fall in the range of 3μH to 20μH. Lower values are chosen to reduce physical size of the inductor. Higher values allow more output current because they reduce peak current seen by the LT1959 switch, which has a 4.5A limit. Higher values also reduce output ripple voltage, and reduce core loss. Graphs in the Typical Performance Characteristics section show maximum output load current versus inductor size and input voltage.

When choosing an inductor you might have to consider maximum load current, core and copper losses, allowable component height, output voltage ripple, EMI, fault current in the inductor, saturation, and of course, cost. The following procedure is suggested as a way of handling these somewhat complicated and conflicting requirements.

1. Choose a value in microhenries from the graphs of maximum load current and core loss. Choosing a small inductor with lighter loads may result in discontinuous mode of operation, but the LT1959 is designed to work well in either mode. Keep in mind that lower core loss means higher cost, at least for closed core geometries like toroids. The core loss graphs show absolute loss for a 3.3V output, so actual percent losses must be calculated for each situation.

Assume that the average inductor current is equal to load current and decide whether or not the inductor must withstand continuous fault conditions. If maximum load current is 0.5A, for instance, a 0.5A inductor may not survive a continuous 4.5A overload condition. Dead shorts will actually be more gentle on the inductor because the LT1959 has foldback current limiting.

2. Calculate peak inductor current at full load current to ensure that the inductor will not saturate. Peak current can be significantly higher than output current, especially with smaller inductors and lighter loads, so don't omit this step. Powdered iron cores are forgiving because they saturate softly, whereas ferrite cores

saturate abruptly. Other core materials fall in between somewhere. The following formula assumes continuous mode of operation, but it errs only slightly on the high side for discontinuous mode, so it can be used for all conditions.

$$I_{\text{PEAK}} = I_{\text{OUT}} + \frac{V_{\text{OUT}}(V_{\text{IN}} - V_{\text{OUT}})}{2(f)(L)(V_{\text{IN}})}$$

V_{IN} = Maximum input voltage

f = Switching frequency, 500kHz

3. Decide if the design can tolerate an "open" core geometry like a rod or barrel, which have high magnetic field radiation, or whether it needs a closed core like a toroid to prevent EMI problems. One would not want an open core next to a magnetic storage media, for instance! This is a tough decision because the rods or barrels are temptingly cheap and small and there are no helpful guidelines to calculate when the magnetic field radiation will be a problem.
4. Start shopping for an inductor (see representative surface mount units in Table 2) which meets the requirements of core shape, peak current (to avoid saturation), average current (to limit heating), and fault current (if the inductor gets too hot, wire insulation will melt and cause turn-to-turn shorts). Keep in mind that all good things like high efficiency, low profile, and high temperature operation will increase cost, sometimes dramatically. Get a quote on the cheapest unit first to calibrate yourself on price, then ask for what you really want.
5. After making an initial choice, consider the secondary things like output voltage ripple, second sourcing, etc. Use the experts in the Linear Technology's applications department if you feel uncertain about the final choice. They have experience with a wide range of inductor types and can tell you about the latest developments in low profile, surface mounting, etc.

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Table 2

VENDOR/ PART NO.	VALUE (μ H)	DC (Amps)	CORE TYPE	SERIES RESIS- TANCE(Ω)	CORE MATER- IAL	HEIGHT (mm)
Coiltronics						
CTX2-1	2	4.1	Tor	0.011	KM μ	4.2
CTX5-4	5	4.4	Tor	0.019	KM μ	6.4
CTX8-4	8	3.5	Tor	0.020	KM μ	6.4
CTX2-1P	2	3.4	Tor	0.014	52	4.2
CTX2-3P	2	4.6	Tor	0.012	52	4.8
CTX5-4P	5	3.3	Tor	0.027	52	6.4
Sumida						
CDRH125	10	4.0	SC	0.025	Fer	6
CDRH125	12	3.5	SC	0.027	Fer	6
CDRH125	15	3.3	SC	0.030	Fer	6
CDRH125	18	3.0	SC	0.034	Fer	6
Coilcraft						
DT3316-222	2.2	5	SC	0.035	Fer	5.1
DT3316-332	3.3	5	SC	0.040	Fer	5.1
DT3316-472	4.7	3	SC	0.045	Fer	5.1
Pulse						
PE-53650	4	4.8	Tor	0.017	52	9.1
PE-53651	5	5.4	Tor	0.018	52	9.1
PE-53652	9	5.5	Tor	0.022	52	10
PE-53653	16	5.1	Tor	0.032	52	10
Dale						
IHSM-4825	2.7	5.1	Open	0.034	Fer	5.6
IHSM-4825	4.7	4.0	Open	0.047	Fer	5.6
IHSM-5832	10	4.3	Open	0.053	Fer	7.1
IHSM-5832	15	3.5	Open	0.078	Fer	7.1
IHSM-7832	22	3.8	Open	0.054	Fer	7.1

Tor = Toroid

SC = Semiclosed geometry

Fer = Ferrite core material

52 = Type 52 powdered iron core material

KM μ = Kool M μ ®

Output Capacitor

The output capacitor is normally chosen by its Effective Series Resistance (ESR), because this is what determines output ripple voltage. At 500kHz, any polarized capacitor is essentially resistive. To get low ESR takes *volume*, so physically smaller capacitors have high ESR. The ESR

range for typical LT1959 applications is 0.05 Ω to 0.2 Ω . A typical output capacitor is an AVX type TPS, 100 μ F at 10V, with a guaranteed ESR less than 0.1 Ω . This is a “D” size surface mount solid tantalum capacitor. TPS capacitors are specially constructed and tested for low ESR, so they give the lowest ESR for a given volume. The value in microfarads is not particularly critical, and values from 22 μ F to greater than 500 μ F work well, but you cannot cheat mother nature on ESR. If you find a tiny 22 μ F solid tantalum capacitor, it will have high ESR, and output ripple voltage will be terrible. Table 3 shows some typical solid tantalum surface mount capacitors.

Table 3. Surface Mount Solid Tantalum Capacitor ESR and Ripple Current

E Case Size	ESR (Max., Ω)	Ripple Current (A)
AVX TPS, Sprague 593D	0.1 to 0.3	0.7 to 1.1
AVX TAJ	0.7 to 0.9	0.4
D Case Size		
AVX TPS, Sprague 593D	0.1 to 0.3	0.7 to 1.1
C Case Size		
AVX TPS	0.2 (typ)	0.5 (typ)

Many engineers have heard that solid tantalum capacitors are prone to failure if they undergo high surge currents. This is historically true, and type TPS capacitors are specially tested for surge capability, but surge ruggedness is not a critical issue with the *output* capacitor. Solid tantalum capacitors fail during very high *turn-on* surges, which do not occur at the output of regulators. High *discharge* surges, such as when the regulator output is dead shorted, do not harm the capacitors.

Unlike the input capacitor, RMS ripple current in the output capacitor is normally low enough that ripple current rating is not an issue. The current waveform is triangular with a typical value of 200mA_{RMS}. The formula to calculate this is:

Output Capacitor Ripple Current (RMS):

$$I_{\text{RIPPLE(RMS)}} = \frac{0.29(V_{\text{OUT}})(V_{\text{IN}} - V_{\text{OUT}})}{(L)(f)(V_{\text{IN}})}$$

Kool M μ is a registered trademark of Magnetics, Inc.

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Ceramic Capacitors

Higher value, lower cost ceramic capacitors are now available in smaller case sizes. These are ideal for input bypassing because of their high ripple rating and tolerance to turn-on surges. As output capacitors, caution must be used. Solid tantalum capacitor's ESR generates a loop "zero" at 5kHz to 50kHz that is beneficial in giving acceptable loop phase margin. Ceramic capacitors remain capacitive to beyond 300kHz and usually resonate with their ESL before ESR becomes effective. When using ceramic output capacitors, the loop compensation pole frequency must be reduced by a typical factor of 10.

OUTPUT RIPPLE VOLTAGE

Figure 3 shows a typical output ripple voltage waveform for the LT1959. Ripple voltage is determined by the high frequency impedance of the output capacitor, and ripple current through the inductor. Peak-to-peak ripple current through the inductor into the output capacitor is:

$$I_{P-P} = \frac{(V_{OUT})(V_{IN} - V_{OUT})}{(V_{IN})(L)(f)}$$

For high frequency switchers, the sum of ripple current slew rates may also be relevant and can be calculated from:

$$\Sigma \frac{dI}{dt} = \frac{V_{IN}}{L}$$

Peak-to-peak output ripple voltage is the sum of a *triwave* created by peak-to-peak ripple current times ESR, and a *square* wave created by parasitic inductance (ESL) and ripple current slew rate. Capacitive reactance is assumed to be small compared to ESR or ESL.

$$V_{RIPPLE} = (I_{P-P})(ESR) + (ESL) \Sigma \frac{dI}{dt}$$

Example: with $V_{IN}=10V$, $V_{OUT}=5V$, $L=10\mu H$, $ESR=0.1\Omega$, $ESL=10nH$:

$$I_{P-P} = \frac{(5)(10-5)}{(10)(10 \cdot 10^{-6})(500 \cdot 10^3)} = 0.5A$$

$$\Sigma \frac{dI}{dt} = \frac{10}{10 \cdot 10^{-6}} = 10^6$$

$$V_{RIPPLE} = (0.5A)(0.1) + (10 \cdot 10^{-9})(10^6)$$

$$= 0.05 + 0.01 = 60mV_{P-P}$$

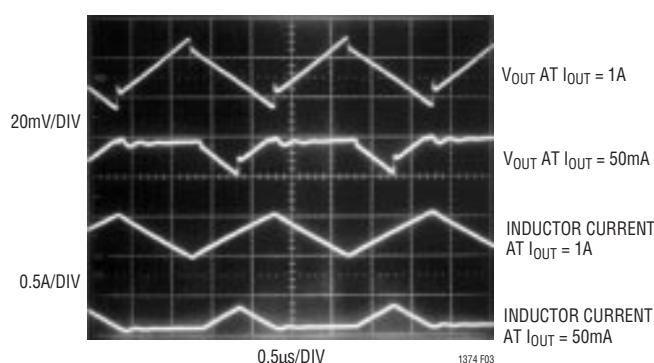


Figure 3. LT1959 Ripple Voltage Waveform

CATCH DIODE

The suggested catch diode (D1) is a 1N5821 Schottky, or its Motorola equivalent, MBR330. It is rated at 3A average forward current and 30V reverse voltage. Typical forward voltage is 0.5V at 3A. The diode conducts current only during switch off time. Peak reverse voltage is equal to regulator input voltage. Average forward current in normal operation can be calculated from:

$$I_{D(AVG)} = \frac{I_{OUT}(V_{IN} - V_{OUT})}{V_{IN}}$$

This formula will not yield values higher than 3A with maximum load current of 4.25A unless the ratio of input to output voltage exceeds 3.4:1. The only reason to consider a larger diode is the worst-case condition of a high input voltage and *overloaded* (not shorted) output. Under short-circuit conditions, foldback current limit will reduce diode current to less than 2.6A, but if the output is overloaded

APPLICATIONS INFORMATION

and does not fall to less than 1/3 of nominal output voltage, foldback will not take effect. With the overloaded condition, output current will increase to a typical value of 5.7A, determined by peak switch current limit of 6A. With $V_{IN} = 15V$, $V_{OUT} = 4V$ (5V overloaded) and $I_{OUT} = 5.7A$:

$$I_{D(AVG)} = \frac{5.7(15 - 4)}{15} = 4.18A$$

This is safe for short periods of time, but it would be prudent to check with the diode manufacturer if continuous operation under these conditions must be tolerated.

BOOST PIN CONSIDERATIONS

For most applications, the boost components are a 0.27 μF capacitor and a 1N914 or 1N4148 diode. The anode is connected to the regulated output voltage and this generates a voltage across the boost capacitor nearly identical to the regulated output. In certain applications, the anode may instead be connected to the unregulated input voltage. This could be necessary if the regulated output voltage is very low (< 3V) or if the input voltage is less than 5V. Efficiency is not affected by the capacitor value, but the capacitor should have an ESR of less than 1 Ω to ensure that it can be recharged fully under the worst-case condition of minimum input voltage. Almost any type of film or ceramic capacitor will work fine.

For nearly all applications, a 0.27 μF boost capacitor works just fine, but for the curious, more details are provided here. The size of the boost capacitor is determined by switch drive current requirements. During switch on time, drain current on the capacitor is approximately $I_{OUT}/50$. At peak load current of 4.25A, this gives a total drain of 85mA. Capacitor ripple voltage is equal to the product of on time and drain current divided by capacitor value; $\Delta V = (t_{ON})(85mA/C)$. To keep capacitor ripple voltage to less than 0.6V (a slightly arbitrary number) at the worst-case condition of $t_{ON} = 1.8\mu s$, the capacitor needs to be 0.27 μF . Boost capacitor ripple voltage is not a critical parameter, but if the minimum voltage across the capacitor drops to less than 3V, the power switch may not saturate fully and efficiency will drop. An *approximate* formula for absolute minimum capacitor value is:

$$C_{MIN} = \frac{(I_{OUT}/50)(V_{OUT}/V_{IN})}{(f)(V_{OUT} - 2.8V)}$$

f = Switching frequency

V_{OUT} = Regulated output voltage

V_{IN} = Minimum input voltage

This formula can yield capacitor values substantially less than 0.27 μF , but it should be used with caution since it does not take into account secondary factors such as capacitor series resistance, capacitance shift with temperature and output overload.

SHUTDOWN FUNCTION AND UNDERVOLTAGE LOCKOUT

Figure 4 shows how to add undervoltage lockout (UVLO) to the LT1959. Typically, ULVO is used in situations where the input supply is *current limited*, or has a relatively high source resistance. A switching regulator draws constant power from the source, so source current increases as source voltage drops. This looks like a negative resistance load to the source and can cause the source to current limit or latch low under low source voltage conditions. ULVO prevents the regulator from operating at source voltages where these problems might occur.

Threshold voltage for lockout is about 2.38V. A 3.5 μA bias current flows *out* of the pin at threshold. This internally generated current is used to force a default high state on the shutdown pin if the pin is left open. When low shutdown current is not an issue, the error due to this current can be minimized by making R_{LO} 10k or less. If shutdown current is an issue, R_{LO} can be raised to 100k, but the error due to initial bias current and changes with temperature should be considered.

$$R_{LO} = 10k \text{ to } 100k \text{ (25k suggested)}$$

$$R_{HI} = \frac{R_{LO}(V_{IN} - 2.38V)}{2.38V - R_{LO}(3.5\mu A)}$$

V_{IN} = Minimum input voltage

2 4-quadrant chopper used in railway traction

The reversible structure of this type of chopper allows for its use in controlling direct current machines in traction, where reversibility is most interesting to exploit. It is used for all power ranges, with the example given here being that of a tramway.

The tramway is modeled by a “car” powered by a DC machine, itself controlled by a reversible chopper powered by a perfect reversible DC source (catenary type). The simplified diagram of the device is shown in the following figure and corresponds to the described quantities.

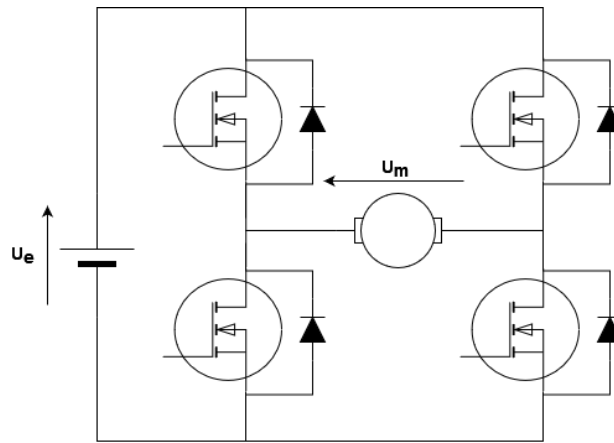


Figure 2.1: 4 Quadrant chopper

The chopper is powered by a constant voltage source with a value of $U_E = 1500V$. It consists of two independent bridge arms controlled at a fixed frequency of $F_d = 2kHz$. The duty cycles are denoted as follows: α_1 for T_{11} and α_2 for T_{21} , with the control of each arm being complementary. We assume perfect components, diodes, and transistors. Between the two bridge arms, the following control relationship is maintained: $\alpha_1 + \alpha_2 = 1$.

The direct current motor is conventionally modeled by its electromotive force (EMF) E , its armature resistance R , and its armature inductance L ; its excitation is kept constant by another device not studied here. The torque constant (equal to the EMF constant) is denoted as k_φ , and the numerical values are as follows:

$R = 0.2\Omega$; $L = 25mH$; $k_\varphi = 7.64Nm/A$; nominal speed $N_n = 1500rpm$.

The tramway car constitutes the mechanical load and presents, with respect to the motor shaft, a moment of inertia $J = 200kg.m^2$. Depending on the trajectory, it also “returns” to this motor shaft a resisting torque denoted as C_v .

1. Total complementary control

- 1.1 Quickly represent the shape of the motor voltage $u_m(t)$ for any duty cycle α_1 , first less than 0.5 and then greater than 0.5.
- 1.2 Calculate the average value of this voltage, denoted as U_m , as a function of α_1 .

2. Functional description

In this part, we neglect the current ripple in the motor. In addition, a very fast current control compared to the mechanical time constants allows us to impose the desired value on the motor armature current, denoted as I_M , which is constant.

- 2.1 Express the average values (at the switching frequency F_d) of the voltage across the motor armature, denoted as U_M , and the current drawn by the chopper from the source, denoted as I_E , as functions of α_1 , U_E , and I_M .
- 2.2 The tramway stops on an **uphill** slope. The weight of the car then exerts a torque on the motor shaft with a value of $C_v = 382Nm$. What is the duty cycle α_{10} required to keep the tramway at a standstill? What is the current I_{E0} drawn from the power supply (still in average value)?
- 2.3 The tramway starts on a level track (neglecting the resisting torque) with a constant acceleration of $5rad/s^2$ until it reaches the nominal motor speed.
 - i. What is the current I_m in the motor?
 - ii. Plot the variation of the duty cycle α_1 during this acceleration phase.
 - iii. Plot the variation of the current I_E drawn from the power supply during this acceleration phase.
 - iv. Calculate the acceleration time Δt_a .
- 2.4 The tramway travels on a **downhill** trajectory at a constant speed, with the motor running at its nominal speed. The weight exerts a torque $C_v = 382Nm$. What is the current I_{Ed} drawn from the power supply? What is the value of the duty cycle α_{1d} that allows for this constant speed operation?
- 2.5 Emergency braking downhill: during the previous operation (1.3), the driver requests emergency braking characterized by controlling the armature current to the maximum value tolerable by the chopper and the machine, $I_{\max} = 250A$. We assume that this current is imposed quasi-instantaneously and remains present during the deceleration phase, with the torque related to the weight of the car remaining unchanged at $382Nm$.
 - i. Calculate and represent the time evolution of the motor speed $\Omega(t)$, which represents the tramway speed.
 - ii. Calculate the deceleration time until stop, denoted as Δt_a .
 - iii. Determine the evolution of the duty cycle $\alpha_1(t)$ during this phase, specifying its value at the beginning of braking and just before the vehicle comes to a stop.

- iv. Calculate the total energy E_C returned to the source during this braking operation.
- 2.6 Represent the operating quadrants of the chopper used during the different phases of the tramway operation.

3 Autonomous electricity production for isolated site

Note: The devices and sizing presented below are a simplified approach to the problem.

The context is that of an isolated site that we want to power using photovoltaic (PV) panels and storage batteries.

The site consumes energy continuously with a “base” power and consumption peaks throughout the day.

To maintain operation during the night, energy storage by batteries is associated with the PV panels.

The energy supply is provided in alternating current, with a service voltage of 230 VAC, 50 Hz generated by a single-phase inverter.

The general architecture of the system is as follows:

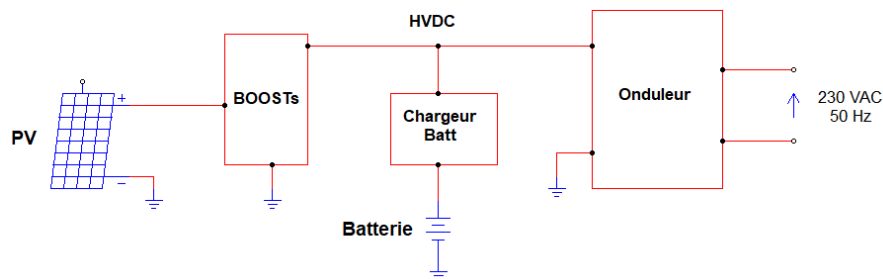


Figure 3.1: Architecture globale

Main data

Installation:

- Base power of the installation: 300 W
- Peak power consumption: 1500 W
- Duration of power peaks: 3×30 min between 11:00 and 15:00.

PV panels:

- The PV panels are oriented to the south, with an inclination of 35°. The installation is located in the Alps, south of Briançon, in a geographical area where the average annual irradiation is 1500 kWh/m² (taking into account shading).
- The datasheet of a PV panel provides the following information:

ELECTRICAL CHARACTERISTICS

PERFORMANCE AT STANDARD TEST CONDITION (STC: 1000W/m², 25°C, AM1.5)

Module Series	TPS105S-250W
Maximum Power at STC (Pmax)	250W (+3%)
Short Circuit Current (Isc)	6.80A
Open Circuit Voltage (Voc)	45.67V
Maximum Power Current (Imp)	6.42A
Maximum Power Voltage (Vmp)	39.01V

MECHANICAL SPECIFICATION

Cell Type	MONO crystalline 182×91mm
Number of Cells	66pcs
Dimensions	1133×1060×30mm
Front Glass	3.2mm Low iron tempered glass
Frame	Anodized aluminum alloy
Connector	MC4 PLUG
Output Cables	Length 0.9m

- The overall efficiency of the PV panels is 20% (taking into account the ratio [cell area / total area]) and it is considered that the overall efficiency of the energy conversion chain is 90% (including converters, line losses, etc.).

Preliminaries

Determine:

1. The energy consumed by the installation per day and per year.
2. The required PV panel area to supply the system, taking into account the efficiencies.
3. The minimum number of PV panels required.

3.1 BOOST Converters

It is decided to implement **12 PV panels**, grouped in two series-connected PV panel groups. Each PV panel group is connected to an input of a BOOST DC-DC converter. Thus, the developed structure of the "BOOST" in Figure 1 is as follows:

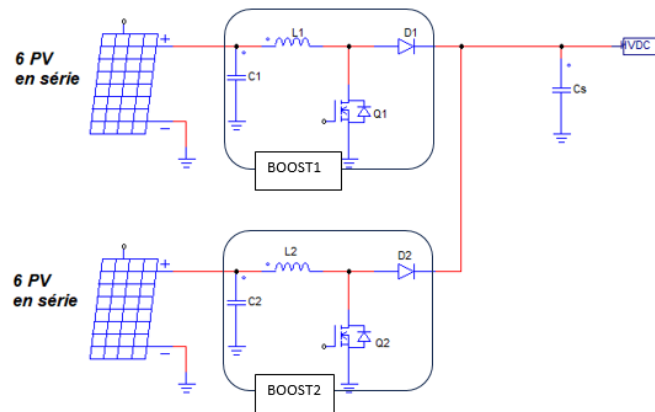


Figure 3.2: Architecture Boost

It is assumed that all PV panels are uniformly illuminated so that each PV panel group provides half of the instantaneous power consumed.

The HVDC output voltage is regulated at **500 V** by control loops acting on the duty cycles of transistors Q1 and Q2.

The switching frequency of Q1 and Q2 is fixed and equal to **Fd = 50 kHz**.

For now, we are only interested in BOOST 1 and we assume that all components are ideal. Considering that:

- The BOOST input voltage (on C1) and HVDC are rigorously constant over one switching period.
 - The ripple current ΔI in L1 represents 20% of the average input current.
 - We are in continuous conduction mode.
1. Represent the currents in L1, Q1, and D1, as well as the voltages across them, over two switching periods and for a duty cycle of 50%.
 2. Determine the expression of HVDC as a function of the input voltage, denoted V_{pv} , and the duty cycle α . Similarly, express the relationship between the input current I_{pv} (average current in L1) and the output current I_{HVDC} .
 3. Determine literally:
 - 3.1 The maximum voltage applied to the terminals of Q1 and D1.
 - 3.2 The average and maximum values of the currents in L1, Q1, and D1 as a function of I_{pv} , ΔI , and the duty cycle.
 - 3.3 Summarize these results in a constraints table (useful for component selection).

Now, consider both BOOST converters operating and sharing the power equally. The environmental conditions are STC conditions ($1000W/m^2$, $25^\circ C$).

We also consider an operating sequence in which (first extreme case):

- The power consumed by the load is maximum.
 - The battery is not fully charged and consumes the power difference between the power consumed by the load and the power supplied by the PV panels, which corresponds to the total "peak" power.
4. Determine:
 - 4.1 The total power P_{cT} that the PV panels can provide considering the irradiation.
 - 4.2 The power flowing through each BOOST converter.
 - 4.3 The power consumed by the battery.
 5. Under these operating conditions, for each of the two BOOST converters, determine:
 - 5.1 The input voltage V_{pv} .

- 5.2 The input current I_{pv} .
- 5.3 The control duty cycle of the transistors.
- 5.4 The value of L for which ΔI represents 20% of I_{pv} .
- 5.5 The constraints applied to the power components Q and D (maximum voltage, average and maximum current).

Now, assume that the power consumed by the load is the minimum power and the battery is fully charged (second extreme case).

- 6. Suppose that the conditions are still STC conditions and that the output voltage of a PV panel is 44 V.
 - 6.1 Determine the power supplied by each group of PV panels and the input current of each BOOST converter.
 - 6.2 Are the BOOST converters still operating in continuous conduction mode?
 - 6.3 Represent the input current in L and the voltage across it.
 - 6.4 Express the input current I_{pv} as a function of V_{pv} , $HVDC$, L , F_d , and the transistor conduction time t_{on} .
 - 6.5 Determine the new value of the duty cycle to maintain HVDC at 500 V.

3.2 Battery Charger (Half-Bridge)

To provide electricity supply during the night, a battery pack is installed.

The power structure associated with the batteries is that of a reversible converter of the “Half Bridge” type (2-quadrant chopper). This converter allows powering the installation from the pack during the night and recharging the pack from the PV panels during the day.

The following diagram shows this structure:

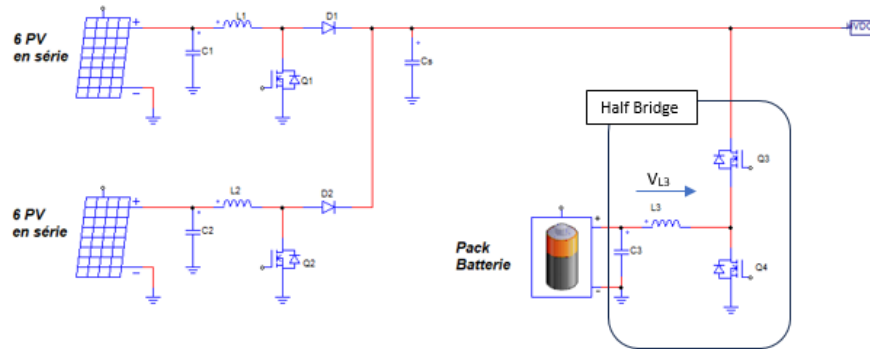


Figure 3.3: 2 Quadrant Chopper Architecture

In all cases, the control loops of the BOOST and Half-Bridge converters maintain **HVDC = 500 V**.

The battery pack voltage is assumed to be constant and equal to **Ubatt = 120 V** (38 elements of 3.2 V / LiFePo4 technology).

A battery current limiting loop included in the BMS (Battery Management System) limits it to $I_{batt_max} = 10 A$.

Transistors Q3 and Q4 are driven in complementary control at a fixed frequency of 50 kHz. Dead times are neglected.

Hereafter, the duty cycle refers to the ratio of the conduction time of Q3 to the switching period.

1. For a given duty cycle (0.3 for example) and in continuous conduction mode, represent, for charge currents ($I_{L3} > 0$) and battery discharge currents of the same absolute values:
 - 1.1 The currents in L3, Q3, and Q4.
 - 1.2 The voltage across L3.
 - 1.3 Indicate the directions of current flow in L3.
2. Express the relationship between the battery voltage and HVDC as a function of the control duty cycle of Q3 denoted α_{HB} (in continuous conduction mode).

Initially, we are in the daytime while the pack is being charged in a configuration corresponding to question 4 of TD 3.1. In this configuration, the battery charging current is I_{batt_max} .

3. Is the Half-Bridge converter operating in “Buck” or “Boost” mode? What is the purpose of C3?
4. We want the ripple current in L3 to represent 30% of its average current (i.e., 3 A). Determine:

4.1 The control duty cycle of Q3.

4.2 The value to be given to L3.

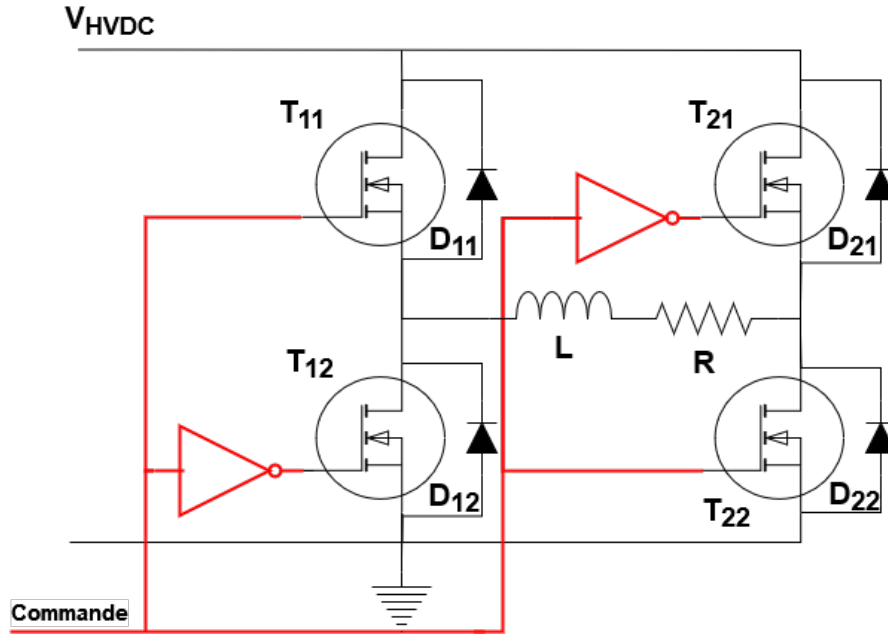
5. For this operating point, determine the voltage, maximum current, and average current constraints on Q3 and Q4.

We are still in the daytime, but now in a configuration where the installation calls for its maximum power (1800 W). The weather conditions are such that the PV panels can only provide 1000 W (including Boost efficiencies).

6. What is the power supplied by the battery and what is its current?
7. Is the Half-Bridge operation “Buck” or “Boost”?
8. What is the control duty cycle of Q3? Is the HB converter still operating in continuous conduction mode? Represent the currents in L3, Q3, and Q4 and the voltage V_{L3} indicating the remarkable numerical values. Provide the equivalent diagrams for the two operating phases indicating the directions of current flow.
9. During the night, the battery alone supplies power to the installation and the power consumed is 300 W. We assume that the HB efficiency is unity.
 - 9.1 Determine the value of the control duty cycle of Q3 to maintain HVDC at 500 V.
 - 9.2 Represent the current in L3 indicating its average value.
10. Taking into account a global efficiency of [HB + Inverter] of 90%, what should be the minimum capacity (in Ah) of the pack to maintain autonomy for 16 consecutive hours with a maximum discharge rate of 90% (during this period, the phases at maximum power do not occur).

3.3 Single-phase PWM Inverter on RL Load

Now we focus on the single-phase inverter. The grid is modeled by a series R-L load. The block diagram is as follows:



- The transistors operate in PWM at a switching frequency F_d much higher than the frequency of the output voltage to be reconstructed.
- The duty cycle varies over time and will be denoted as $\alpha(t) = 0.5 + \delta_\alpha \sin(\omega t)$.

Hereafter, we refer to:

- V_0 : the voltage across the $L - R$ combination (inverter output voltage).
- I_R : the current in the R-L load.

The main characteristics of the load are:

- Equivalent circuit: series $R - L$, with $L = 30mH$
- Rated power: 1500 W
- Rated voltage across R : $U_{Rn} = 230V$ (rms value)

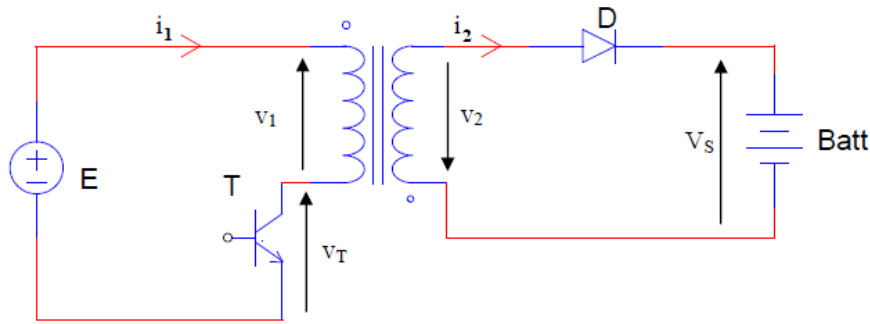
1. Output voltage V_0

- 1.1 Represent the shape of the output voltage for a duty cycle $\alpha > 0.5$ over one switching period.
- 1.2 Express the average value of V_0 over one switching period as a function of the duty cycle α and V_{DC} .

- 1.3 Express the time-varying average value (over one switching period) of the output voltage $\langle V_0 \rangle(t)$ as a function of V_{DC} and δ_α .
- 1.4 Determine the rms value of $\langle V_0 \rangle(t)$ (denoted $V_{0\text{ eff}}$) as a function of V_{HVDC} and δ_α .
- 1.5 Neglecting the current component due to switching, give the expression of the current in R ($I_R(t)$) taking $\langle V_0 \rangle(t)$ as the reference phase.
- 1.6 Express the rms value of the current in R as a function of $V_{0\text{ eff}}$, R , L , and ω .
- 1.7 Express the phase shift φ between $I_R(t)$ and $\langle V_0 \rangle(t)$ as a function of R , L , and ω .
- 1.8 Express the power factor of the RL load as a function of R , L , and ω .
- 1.9 Express the power in R as a function of δ_α , V_{HVDC} , R , and φ .
2. Operation at rated power (1500W).
 As a reminder, the supply voltage V_{HVDC} is the DC bus voltage, $V_{HVDC} = 500V$. The frequency of the reconstructed voltage ($\langle V_0 \rangle(t)$) is **50 Hz**.
 - 2.1 Determine the value of R for this operating point.
 - 2.2 Determine the power factor value.
 - 2.3 Establish the Fresnel diagram relating V_0 , I_R , and U_R .
 - 2.4 Determine the value of the voltage across L .
 - 2.5 Determine the value to be given to $V_{0\text{ eff}}$.
 - 2.6 Determine the value to be given to δ_α to achieve this operating point.
3. Determine the maximum voltage in the blocked state and the maximum current for each inverter switch (the current component at the switching frequency is always neglected).

4 Flyback converter for battery charger

The goal is to use the circuit shown in the following figure to create a 24V battery charger.



Let n_1 and n_2 be the respective numbers of turns in the primary and secondary windings, with the transformation ratio denoted as m . The primary magnetizing inductance is denoted as L_1 , the switching frequency as f , and the duty cycle as α .

The initially discharged battery has a voltage of $V_{S0} = 20V$ across its terminals. The charging process will be stopped when the voltage across the terminals reaches $V_{S1} = 26V$. However, throughout the analysis, this battery voltage, denoted as V_S , will be assumed to be constant on the timescale of the switching.

At the beginning of the battery charging process, it is assumed that the power supply operates in critical mode with a duty cycle of α_0 , and the average current in the battery is I_{S0} .

Numerical values: $E = 300V$; $f = 20kHz$; $\alpha_0 = 0.5$; $I_{S0} = 10A$.

1. Start of charging

- 1.1 Provide the expression of Hopkinson's theorem relating n_1 , i_1 , n_2 , i_2 , and R (reluctance of the ferrite core, inverse of the specific inductance A_L), and φ (elementary flux in the core).
- 1.2 Plot the waveforms of i_1 , i_2 , φ , v_T , and v_1 over one switching period.
- 1.3 Deduce the expression of the output voltage V_S in terms of E , m , and α_0 . Calculate the numerical value of m .
- 1.4 Determine the expression of the average output current I_{S0} in terms of L_1 , α_0 , m , E , and f . Calculate the numerical value of L_1 .
- 1.5 Calculate the energy W_D delivered to the battery per switching cycle. Determine the charging time at constant current for a battery with a "capacity" of 25 Ah.

2. End of charging

During the charging process, the output voltage slowly increases from V_{S0} to V_{S1} while maintaining the duty cycle at α_0 .

- 2.1 How has the conduction mode of the power supply been modified?
- 2.2 Plot the waveforms of i_1 , i_2 , v_T , v_1 and φ over one switching period.
- 2.3 What is the conduction time of the diode D?
- 2.4 What is the energy transferred per cycle now? Determine the average current I_S supplied to the battery.
- 2.5 What value α_1 should be given to the duty cycle to subsequently provide an average charging current of $I_{S1} = I_{S0}/10$?

3. Saturation limit

We are in the critical mode (start of charging).

- 3.1 Determine the expression of the maximum magnetic induction in the ferrite core in terms of E , α , T , n_1 , and A_e (effective area of the core).
- 3.2 Deduce the expression of the minimum value $A_{e \min}$ of the core area to prevent any saturation of the material. For this purpose, let B_{sat} be the induction value near saturation.

4. Winding window area

We are still in the critical mode (start of charging).

- 4.1 Using the results from the first part, determine the RMS value of the current in the transistor $I_{1 \text{ eff}}$ in terms of the output current I_s , m , and α .
- 4.2 Similarly, express the RMS value of the secondary current $I_{2 \text{ eff}}$ in terms of I_s and α .
- 4.3 The chosen current density for both windings made of copper wire is δ . Determine the expression of the total copper area, denoted as $S_{T \text{ Cu}}$, in terms of I_s , δ , n_2 , and α . Compare the primary and secondary copper areas.
- 4.4 Deduce the minimum winding window area, denoted as S_B , if a winding coefficient k_B is chosen ($S_{T \text{ Cu}} = k_B \cdot S_B$).

5. Choice of magnetic core

- 5.1 Based on the previous questions, determine the inequality that the product $A_e \cdot S_B$ of this power supply must satisfy in terms of B_{sat} , P_s , T , k_B , δ , and α .
- 5.2 Calculate the minimum value of this product in mm^4 for the following numerical values:

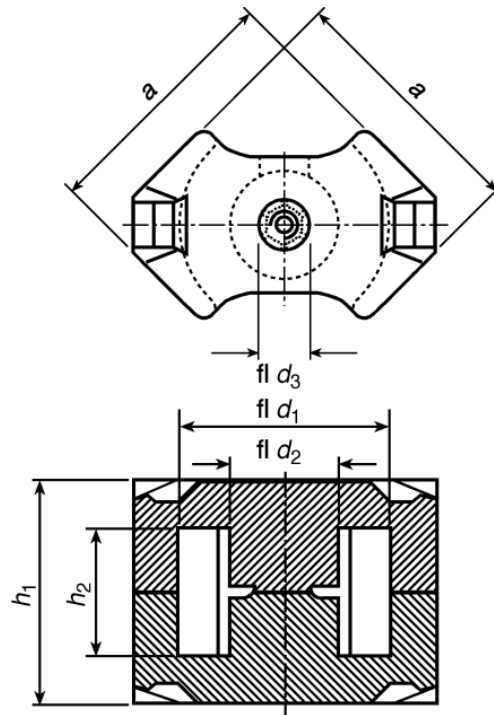
$$P_s = 200W; \alpha = 0.5; k_B = 0.5; B_{sat} = 0.4T; \delta = 5A.\text{mm}^{-2}; f = 20kHz.$$

- 5.3 Choose a core from the following list, for which the manufacturer provides the values of A_e , S_B , and A_L (specific inductance) without air gap, as well as A_L with a 0.5 mm air gap. Note that all these cores are made of the same material, N41.

Core	A_e (mm ²)	S_B (mm ²)	A_L without air gap (nH)	A_L 0.5 mm air gap (nH)
RM6	31	32	3100	—
RM8	55	57	4100	160
RM10	90	93	5500	250
RM12	125	128	6000	300
RM14	170	176	6800	400

- 5.4 For this choice of core, determine the minimum numbers of turns n_1 and n_2 to limit the maximum induction in the ferrite core to B_{sat} . For these calculations, choose the values of the numbers of turns that are “optimal” for the given numerical values.

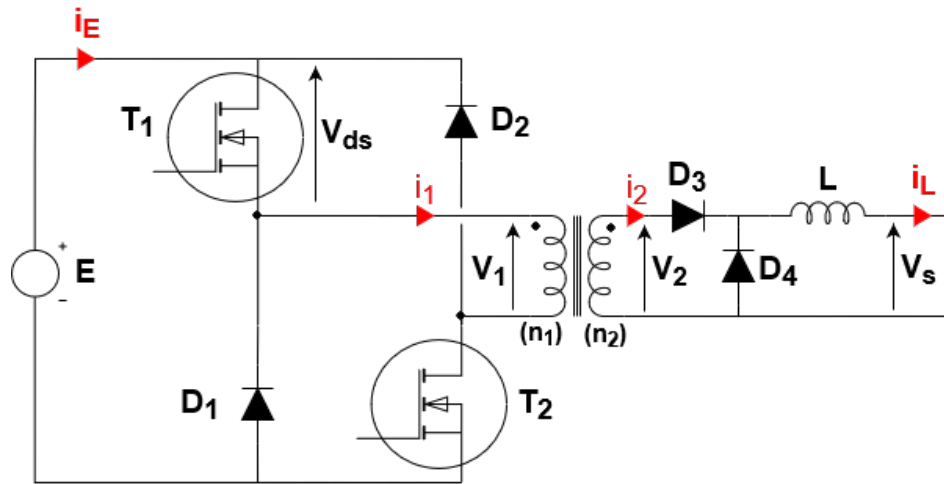
- 5.5 Will an air gap be necessary to achieve the desired inductance L_1 ?



RM- 8- bis RM-14-Kernstze
fr nichtlineare Drosselspulen

5 Direct Switching Power Supply (Asymmetric FORWARD)

The purpose of this problem is the simplified study of the switching converter shown in the following figure:



All semiconductors are assumed to be perfect. The two windings of the transformer are made on the same ferromagnetic core without leakage: all the turns therefore “enclose” the same elementary flux φ , the reluctance of this magnetic circuit being denoted R . The output inductor L is independent of this transformer. All resistances are neglected.

The control of the transistors is identical, carried out at the period T , with a duty cycle denoted α : they are therefore conducting on the interval $[0, \alpha.T]$, blocked on the interval $[\alpha.T, T]$.

The output voltage V_s is assumed to be constant throughout the study and the device will be maintained in total demagnetization at each operating cycle. We are only interested in the continuous conduction mode in the output inductor L , so its current never goes to zero: $i_L > 0$.

1. General operation of the power supply

In this part, we will study the operation of the power supply. We will assume that it operates in **total demagnetization**, meaning that at the end of a period, the magnetic flux in the transformer is zero. We will plot the time curves of φ (magnetic flux in the transformer), v_1 , v_2 , v_{DS} (drain-source voltage of transistor T_1), v_L , i_1 , i_2 , i_L , and i_E .

- 1.1 Considering the directions of the windings shown in the diagram and the directions of measurement of the different currents and voltages:
 - i. Give the expression of the voltages v_1 and v_2 as a function of the flux φ
 - ii. Express the Hopkinson's law relating the currents i_1 and i_2 to the same flux φ using the number of turns and the reluctance R .
- 1.2 We consider the interval $[0, \alpha T]$: we assume that the initial flux at $t = 0$ is zero (total demagnetization)
 - i. Determine the voltage v_1 applied to the primary winding
 - ii. Deduce the voltage v_2 as well as the state of diodes $D3$ and $D4$
 - iii. Give the expression of the voltage v_L across the output inductance in terms of the number of turns, E , and V_S
 - iv. Express the evolution of the flux $\varphi(t)$ during this time interval
 - v. Express the current i_2 in terms of i_L
 - vi. Using the Hopkinson's law and the previous results, determine the expression of the primary current i_1 in terms of i_L and the flux φ using the number of turns and the reluctance R
 - vii. At the instant αT , the end of this first time interval, we assume that the flux has become equal to φ_0 . Deduce the expression of this value of φ_0 in terms of E , n_1 , and αT
- 1.3 We consider the interval $[\alpha T, T]$: all transistors are blocked.
 - i. At $t = \alpha T$, determine the state of diodes $D1$ and $D2$. What are the values of voltages v_1 and v_2 ? Deduce the state of diodes $D3$ and $D4$
 - ii. Deduce the expression of v_L in terms of V_S
 - iii. Express the evolution of the flux $\varphi(t)$ during this time interval
 - iv. At what instant t_0 does the flux become zero? Give a condition on α for the system to operate in total demagnetization
 - v. What happens next until the end of the period T ?
- 1.4 The power supply is intended to produce a constant voltage of 24 V from the perfectly filtered single-phase rectified network, $E = 300V$. The transmitted power is $P_n = 600W$, and the switching frequency is $f = 50kHz$. We choose to operate the power supply around the nominal duty cycle $\alpha = 0.4$.
 - i. Calculate the turns ratio n_2/n_1
 - ii. Calculate the average value I_s of the current i_L
 - iii. Calculate the minimum value to be given to the inductance L to limit the peak-to-peak ripple of the current i_L to 5% of its average value I_s . Keep this value of L for the following calculations

- iv. Deduce the maximum value $I_{1 \text{ max}}$ of the current in the transistor when neglecting the magnetizing current

2. Transformer sizing

- 2.1 To justify the previous assumption, determine the minimum value of the magnetizing inductance L_1 (seen from the primary winding) that makes the maximum no-load current less than $I_{1 \text{ max}}/20$
- 2.2 Neglecting the ripple of the current i_L and the magnetizing current, determine the rms values $I_{1 \text{ eff}}$ and $I_{2 \text{ eff}}$ of the currents i_1 and i_2 in terms of I_s , n_1 , n_2 , and α
- 2.3 Determine, with these same assumptions, the minimum winding area S_B required in terms of current density δ , winding coefficient k_B , I_s , n_2 , and α
- 2.4 Also give the expression of the minimum cross-sectional area A_e of the core that avoids core saturation by maintaining a maximum induction below B_{max}
- 2.5 Deduce the minimal sizing product $A_e.S_B$ of this transformer in terms of the duty cycle, the different characteristic parameters of this power supply, including its output power P_S , and its switching frequency f

